41-Xalqaro olimpiada

1-MASALA. METALL JISM YONIDAGI ZARYAD – TASVIRLAR USULI

Kirish – Tasvirlar usuli

R radiusli metall shar (asoslangan) yaqiniga q nuqtaviy zaryadi joylashtirilgan [1(a)-rasm]. Buning natijasida shar sirtida sirt zaryad taqsimoti induksiyalanadi. Sirt zaryadining elektr maydoni va potensialini hisoblash juda murakkab. Biroq, bu hisoblashni tasvirlar usuli deb ataladigan usul yordamida ancha soddalashtirish mumkin. Bu usulda shar sirtida taqsimlangan zaryad hosil qilgan elektr maydoni va potensialini shar ichida joylashtirilgan bitta q' nuqtaviy zaryadining maydoni va potensiali sifatida ifodalash mumkin (buni isbotlash shart emas). Izoh: Bu tasvir zaryadining q' elektr maydoni faqat shar tashqarisida (shu jumladan uning sirtida) haqiqiy maydonni qayta hosil qiladi.

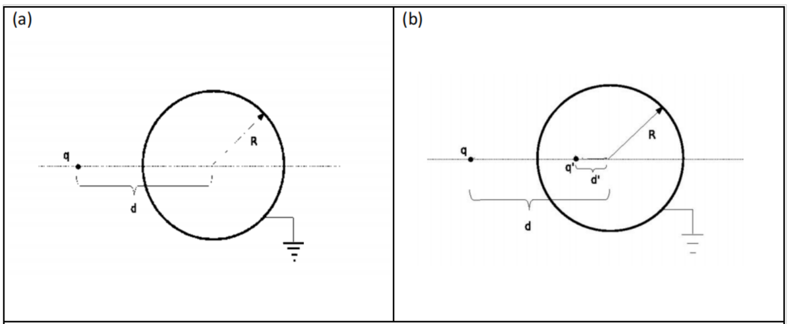

Fig 1. (a) A point charge q in the vicinity of a grounded metallic sphere. (b) The electric field of the charge induced on the sphere can be represented as electric field of an image charge q' .

Figure 1: 1(a)-rasm. Asoslangan metall shar yaqinidagi q nuqtaviy zaryadi. (b) Shar sirtida induksiyalangan zaryadning elektr maydoni q' tasvir zaryadining maydoni sifatida ifodalanishi.

1-topshiriq – Tasvir zaryadi

Masala simmetriyasiga ko‘ra, q' zaryad q nuqtaviy zaryadi va shar markazini birlashtiruvchi to‘g‘ri chiziqda joylashishi kerak [1(b)-rasm].

1. Sferadagi potensialning qiymati qancha? *(0.3 ball)*
2. q' va uning shar markazidan masofasi d' ni q, d, R orqali ifodalang. *(1.9 ball)*
3. q zaryadiga ta’sir qiluvchi kuchning kattaligini toping. Bu kuch itaruvchimi? *(0.5 ball)*

2-topshiriq – Elektrostatik maydonning ekranlanishi

Asoslangan R radiusli metall shar markazidan d masofada q nuqtaviy zaryadi joylashtirilgan. Bizni asoslangan metall shar shar qarama-qarshi tomonidagi A nuqtada elektr maydoniga qanday ta’sir qilishi qiziqtiradi (2-rasm). A nuqta q zaryadi va shar markazini birlashtiruvchi chiziqda yotadi; uning q zaryadidan masofasi r.

1. A nuqtadagi elektr maydon vektorini toping. *(0.6 ball)*

2. Juda katta masofa $r \gg d$ uchun $(1+a)^{-2} \approx 1-2a$ (bunda $a \ll 1$) yaqinlashishidan foydalanib, elektr maydon ifodasini toping. *(0.6 ball)*

3. Asoslangan metall shar q zaryadining maydonini to‘liq ekranlaydigan, ya’ni A nuqtadagi elektr maydoni aynan nolga teng bo‘ladigan d ning qanday chegarasida? *(0.3 ball)*

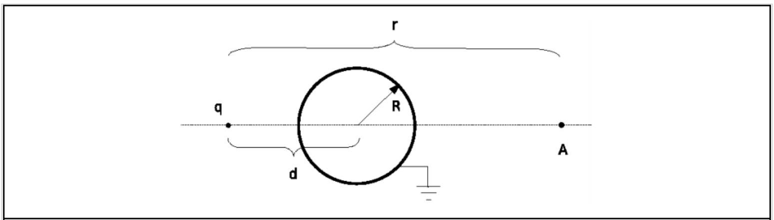

Fig 2. The electric field at point *A* is partially screened by the grounded sphere.

Figure 2: 2-rasm. Asoslangan shar tomonidan A nuqtadagi elektr maydoni qisman ekranlanadi.

3-topshiriq – Asoslangan metall shar elektr maydonidagi kichik tebranishlar

m massali q nuqtaviy zaryadi devorga biriktirilgan l uzunlikdagi ipga osilgan va asoslangan metall shar yaqinida joylashgan. Devorning barcha elektrostatik ta’sirlarini e’tiborga olmang. Nuqtaviy zaryad matematik mayatnik hosil qiladi (3-rasm). Ipning devorga biriktirilgan nuqtasi shar markazidan l masofada. Gravitatsiya ta’sirini e’tiborga olmang.

1. Berilgan α burchak uchun q nuqtaviy zaryadiga ta’sir qiluvchi elektr kuchining kattaligini toping va aniq diagrammada yo‘nalishini ko‘rsating. *(0.8 ball)*

2. Ushbu kuchning ipga perpendikulyar yo‘nalishdagi tashkil etuvchisini l, R, q va α orqali ifodalang. *(0.8 ball)*

3. Mayatnikning kichik tebranishlari chastotasini toping. *(1.0 ball)*

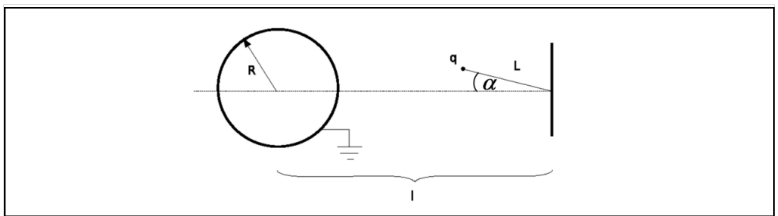

Fig 3. A point charge in the vicinity of a grounded sphere oscillates as a pendulum.

Figure 3: 3-rasm. Asoslangan shar yaqinidagi nuqtaviy zaryad mayatnik sifatida tebranadi.

4-topshiriq – Tizimning elektrostatik energiyasi

Elektr zaryadlari taqsimoti uchun tizimning elektrostatik energiyasini bilish muhim. Bizning masalamizda (1a-rasm) tashqi q zaryadi va sferada induksiyalangan zaryadlar orasida elektrostatik o‘zaro ta’sir, shuningdek induksiyalangan zaryadlarning o‘zaro elektrostatik ta’siri mavjud. Quyidagi elektrostatik energiyalarni q, R, d orqali aniqlang:

1. q zaryadi va sferada induksiyalangan zaryadlar orasidagi o‘zaro ta’sirning elektrostatik energiyasi; *(1.0 ball)*
2. Sferada induksiyalangan zaryadlarning o‘zaro ta’sir elektrostatik energiyasi; *(1.2 ball)*
3. Tizimdagi o‘zaro ta’sirning umumiy elektrostatik energiyasi. *(1.0 ball)*

Ko‘rsatma: Masalani yechishning bir necha usuli bor:

1. Quyidagi integraldan foydalanish mumkin:

$$\int_0^R \frac{xdx}{(x^2 - R^2)^2} = \frac{1}{2}\frac{1}{d^2 - R^2}.$$

2. Yana bir usulda, $\vec{r}_i$ nuqtalarda joylashgan N ta q_i zaryadlari to‘plami uchun elektrostatik energiya barcha juftliklar yig‘indisiga teng:

$$V = \frac{1}{2}\sum_{i=1}^{N}\sum_{\substack{j=1 \\ i\neq j}}^{N} \frac{q_i q_j}{|\vec{r}_i - \vec{r}_j|}.$$

2-MASALA. MO'RI FIZIKASI

Kirish

Yonish gazsimon mahsulotlari T_{Havo} haroratli atmosferaga A ko'ndalang kesim va h balandlikdagi mo'ri orqali chiqariladi (1-rasm). Qattiq modda T_{Tutun} haroratli o'choqda yondiriladi. O'choqda vaqt birligida hosil bo'ladigan gaz hajmi B.

Quyidagilarni faraz qiling:

- o'choqdagi gazlarning tezligi juda kichik (ahamiyatsiz);
- gazlar (tutun) zichligi bir xil harorat va bosimdagi havoning zichligidan farq qilmaydi; o'choqdagi gazlarni ideal deb hisoblash mumkin;
- havo bosimi balandlik bilan gidrostatik qonunga muvofiq o'zgaradi; havo zichligining balandlik bilan o'zgarishi ahamiyatsiz;
- gaz oqimi Bernulli tenglamasini qanoatlantiradi, unga ko'ra oqimning barcha nuqtalarida quyidagi kattalik saqlanadi:

$$\frac{1}{2}\rho v^2(z) + \rho g z + p(z) = \text{const},$$

  bunda ρ – gaz zichligi, $v(z)$ – uning tezligi, $p(z)$ – bosim, z – balandlik;
- gaz zichligining mo'ri bo'ylab o'zgarishi ahamiyatsiz.

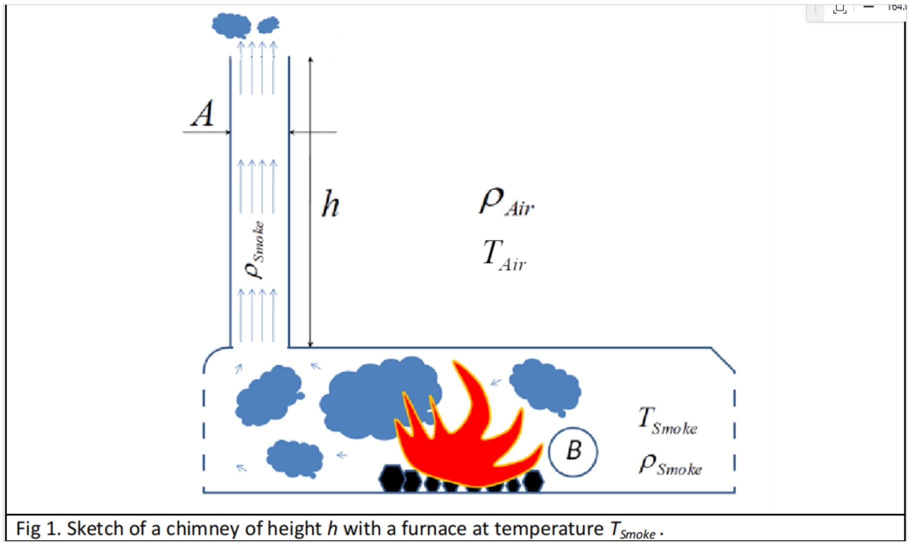

Figure 4: 1-rasm. Balandligi h bo'lgan mo'ri va T_{Tutun} haroratli o'choq.

1-topshiriq

1. Mo'rining samarali ishlashi, ya'ni hosil bo'lgan barcha gazni atmosferaga chiqarib yuborishi uchun zarur bo'lgan minimal balandlik qancha? Javobingizni $B, A, T_{\text{Havo}}, g = 9.81\,\text{m/s}^2, \Delta T = T_{\text{Tutun}} - T_{\text{Havo}}$ orqali ifodalang. **Muhim:** keyingi barcha topshiriqlarda mo'rining balandligi shu minimal balandlik deb qabul qilinsin. *(3.5 ball)*

2. Ikkita mo‘ri xuddi shu maqsadda qurilgan deb faraz qilaylik. Ularning ko‘ndalang kesimlari bir xil, lekin dunyoning turli mintaqalarida ishlashga mo‘ljallangan: biri sovuq mintaqada (atmosferaning o‘rtacha harorati -30 °C), ikkinchisi issiq mintaqada (o‘rtacha harorat 30 °C). O‘choq harorati 400 °C. Sovuq mintaqada ishlashga mo‘ljallangan mo‘rining balandligi 100 m deb hisoblangan. Ikkinchi mo‘rining balandligi qancha? *(0.5 ball)*
3. Gazlar tezligi mo‘ri bo‘ylab qanday o‘zgaradi? Mo‘rining ko‘ndalang kesimi bo‘ylab o‘zgarmas deb faraz qilib, chizma (diagramma) chizing. Gazlarning mo‘riga kirish nuqtasini ko‘rsating. *(0.6 ball)*
4. Gazlar bosimi mo‘ri bo‘ylab qanday o‘zgaradi? *(0.5 ball)*

Quyosh elektr stansiyasi

Mo‘ridagi gaz oqimidan ma’lum turdagi quyosh elektr stansiyasini (quyosh mo‘ri) qurishda foydalanish mumkin. G‘oya 2-rasmda tasvirlangan. Quyosh S maydonli kollektor ostidagi havoni isitadi, kollektor atrofi ochiq bo‘lib, havoning to‘sqinliksiz kirishiga imkon beradi (2-rasm). Isitilgan havo mo‘ri orqali yuqoriga ko‘tarilganda (yo‘g‘on uzluksiz o‘qlar), yangi sovuq havo kollektorga uning atrofidan kiradi (qalin nuqtali o‘qlar), bu esa elektr stansiyasi orqali havoning uzluksiz oqishini ta’minlaydi. Mo‘ri orqali o‘tadigan havo oqimi turbinani harakatga keltirib, elektr energiyasini ishlab chiqaradi. Kollektor gorizontal yuzasining birlik yuzasiga vaqt birligida tushadigan quyosh radiatsiyasi energiyasi G. Faraz qilaylik, bu energiyaning hammasi kollektordagi havoni isitishga sarflanadi (havoning massa issiqlik sig‘imi c va uning haroratga bog‘liqligini e’tiborsiz qoldirish mumkin). Quyosh mo‘rining foydali ish koeffitsienti deb gaz oqimining kinetik energiyasining mo‘riga kirishidan oldin havoni isitishda yutilgan quyosh energiyasiga nisbatini olamiz.

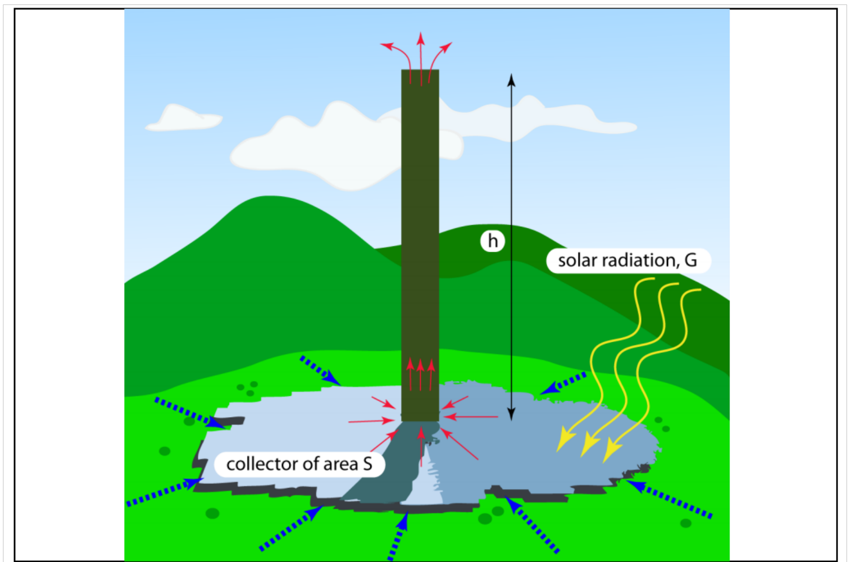

Fig 2. Sketch of a solar power plant.

Figure 5: 2-rasm. Quyosh mo‘rining prinsipial sxemasi.

2-topshiriq

1. Quyosh mo‘ri elektr stansiyasining foydali ish koeffitsienti qancha? *(2.0 ball)*

2. Mo‘ri balandligiga qarab FIKning o‘zgarish diagrammasini chizing. *(0.4 ball)*

Manzanares prototipi

Ispaniyaning Manzanares shahrida qurilgan prototip mo‘ri balandligi 195 m, radiusi 5 m edi. Kollektor aylana shaklida bo‘lib, diametri 244 m. Quyosh mo‘ri prototipining odatdagi ishlash sharoitida havoning solishtirma issiqlik sig‘imi 1012 J/(kg·K), issiq havoning zichligi taxminan 0,9 kg/m^3, atmosferaning odatdagi harorati $T_{\mathrm{Havo}} = 295$ K. Manzanaresda quyoshli kunda gorizontal sirt birligiga to‘g‘ri keladigan quyosh quvvati odatda 150 W/m^2.

3-topshiriq

1. Prototip elektr stansiyasining FIKi qancha? Raqamli bahosini yozing. *(0.3 ball)*
2. Prototipda qancha quvvat ishlab chiqarilishi mumkin? *(0.4 ball)*
3. Oddiy quyoshli kunda (kun davomiyligi 8–10 soat deb oling) stansiya qancha energiya ishlab chiqarishi mumkin? *(0.3 ball)*

4-topshiriq

1. Havoning atrofdagi (sovuq havo) mo‘riga kiradigan (issiq havo) haroratining o‘sishi qancha? Umumiy formulani yozing va prototip mo‘ri uchun baholang. *(1.0 ball)*
2. Tizim orqali havoning massaviy oqim tezligi qancha? *(0.5 ball)*

3-MASALA. ATOM YADROSINING SODDA MODELI

Kirish

Atom yadrolari kvant ob'yektlari bo'lsa-da, ularning asosiy xossalari (masalan, radius yoki bog'lanish energiyasi) uchun bir qator fenomenologik qonuniyatlar oddiy farazlardan kelib chiqadi: (i) yadrolar nuklonlardan (proton va neytronlardan) tuzilgan; (ii) bu nuklonlarni bir-biriga bog'lab turuvchi kuchli yadro ta'siri juda qisqa masofali (faqat qo'shni nuklonlar orasida ta'sir qiladi); (iii) berilgan yadrodagi protonlar soni Z neytronlar soni N ga taxminan teng, ya'ni $Z \approx N \approx A/2$, bunda A – nuklonlarning umumiy soni ($A \gg 1$). Muhim: Quyidagi 1–4-topshiriqlarda ushbu farazlardan foydalaning.

1-topshiriq – Atom yadrosi nuklonlarning zich joylashgan tizimi sifatida

Oddiy modelda atom yadrosini bir-biriga zich joylashgan nuklonlardan iborat shar deb qarash mumkin [1(a)-rasm], bunda nuklonlar radiusi $r_N = 0,85$ fm (1 fm $= 10^{-15}$ m) bo'lgan qattiq sharchalardir. Yadro kuchi faqat ikkita nuklon bir-biriga tegib turganida mavjud. Yadro hajmi V barcha nuklonlar hajmi AV_N dan katta, bu yerda $V_N = \frac{4}{3}\pi r_N^3$. $f = AV_N/V$ nisbat zichlik koeffitsienti (packing factor) deb ataladi va yadro moddasi bilan to'ldirilgan foydali hajm ulushini beradi.

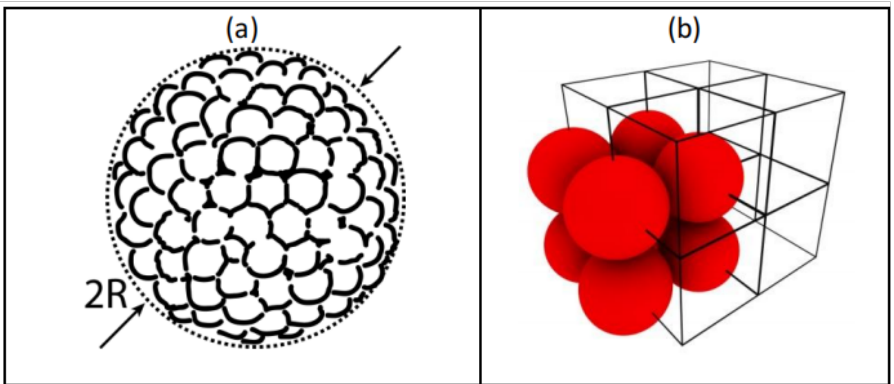

Fig. 1. (a) An atomic nucleus as a ball of closely packed nucleons. (b) The SC packing.

Figure 6: 1-rasm. (a) Atom yadrosi – zich joylashgan nuklonlar shari. (b) Oddiy kubik (SC) joylashish.

1. Agar nuklonlar "oddiy kubik" (SC) kristall tizimida, ya'ni har bir nuklon cheksiz kubik panjaraning tugun nuqtasida joylashgan bo'lsa, f zichlik koeffitsienti qancha bo'ladi? *(0.3 ball)*

   Muhim: Keyingi barcha topshiriqlarda yadrolar uchun haqiqiy zichlik koeffitsienti 1a-topshiriqdagi qiymatga teng deb faraz qiling. Agar uni hisoblay olmasangiz, keyingi topshiriqlarda $f = 1/2$ dan foydalaning.

2. A nuklonli yadro uchun o'rtacha massa zichligi ρ_m, zaryad zichligi ρ_c va radius R ni baholang. Nuklonning o'rtacha massasi $1,67 \cdot 10^{-27}$ kg. *(1.0 ball)*

2-topshiriq – Atom yadrolarining bog‘lanish energiyasi – hajm va sirt hadlari

Yadroning bog‘lanish energiyasi deb uni alohida nuklonlarga ajratish uchun zarur bo‘lgan energiyaga aytiladi va u asosan har bir nuklonning qo‘shnilari bilan tortishish yadro kuchidan kelib chiqadi. Agar nuklon yadro sirtida bo‘lmasa, u umumiy bog‘lanish energiyasiga $a_v = 15,8$ MeV (1 MeV $= 1,602 \cdot 10^{-13}$ J) hissa qo‘shadi. Bitta sirt nuklonining bog‘lanish energiyasiga qo‘shgan hissasi taxminan $a_v/2$ ga teng. A nuklonli yadroning bog‘lanish energiyasi E_b ni A, a_v va f orqali, sirt tuzatishini ham qo‘shib ifodalang. *(1.9 ball)*

3-topshiriq – Bog‘lanish energiyasiga elektrostatik (Kulon) effektlar

Bir jinsli zaryadlangan shar (radiusi R va umumiy zaryadi Q_0) ning elektrostatik energiyasi

$$U_c = \frac{3Q_0^2}{20\pi\epsilon_0 R},$$

bunda $\epsilon_0 = 8,85 \cdot 10^{-12}$ $\mathrm{C^2/(N \cdot m^2)}$.

1. Ushbu formulani yadroga qo‘llang. Yadroda har bir proton o‘ziga emas, balki qolgan protonlarga nisbatan Kulon kuchi ta’sirida bo‘ladi. Olingan formulada Z^2 ni $Z(Z-1)$ ga almashtirish orqali buni hisobga olish mumkin. Keyingi topshiriqlarda ushbu tuzatishdan foydalaning. *(0.4 ball)*
2. Bog‘lanish energiyasining to‘liq formulasini yozing (asosiy – hajm hadi, sirt tuzatish hadi va olingan elektrostatik tuzatish). *(0.3 ball)*

4-topshiriq – Og‘ir yadrolarning bo‘linishi

Bo‘linish – bu yadroning kichikroq qismlarga (yengilroq yadrolarga) parchalanishi jarayoni. Faraz qilaylik, A nuklonli yadro 2-rasmda ko‘rsatilganidek, faqat ikkita teng qismga bo‘linsin.

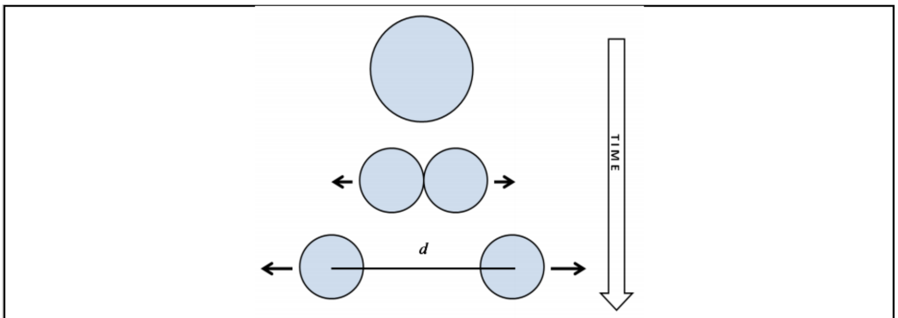

Fig. 2. A schematic description of nuclear fission in our model.

Figure 7: 2-rasm. Modelimizda yadro bo‘linishining sxematik tasviri.

1. Ikki yengil yadro markazlari orasidagi masofa $d \geq 2R(A/2)$ (bu yerda $R(A/2)$ – ularning radiusi) bo‘lganda, bo‘linish mahsulotlarining umumiy kinetik energiyasi E_{kin} ni hisoblang. Katta yadro dastlab tinch holatda edi. *(1.3 ball)*

2. $d = 2R(A/2)$ deb olib, a) qismida olingan E_{kin} ifodasini $A = 100, 150, 200$ va 250 uchun hisoblang (natijalarni MeV da ifodalang). Yuqorida tasvirlangan modelda bo‘linish qanday A qiymatlarida mumkin? *(1.0 ball)*

5-topshiriq – Transfer reaksiyalari

1. Zamonaviy fizikada yadrolarning energetikasi va ularning reaksiyalari massalar tili bilan ifodalanadi. Masalan, tinch holatdagi yadro asosiy holatda m_0 massaga ega bo‘lsa, qo‘zg‘algan holatda uning massasi $m = m_0 + E_{\text{exc}}/c^2$ bo‘ladi, bu yerda E_{exc} – qo‘zg‘alish energiyasi. $^{16}\text{O} +^{54}\text{Fe} \rightarrow^{12}\text{C} +^{58}\text{Ni}$ reaksiyasi "transfer reaksiyalari"ga misol bo‘lib, unda bir yadroning bir qismi ("klaster") ikkinchi yadroga o‘tadi (3-rasm). Bizning misolda o‘tkazilayotgan qism ^{4}He-klaster (α-zarracha). Transfer reaksiyalari eng katta ehtimollik bilan, agar snaryadga o‘xshash reaksiya mahsulotining (bizning holda ^{12}C) tezligi kattaligi va yo‘nalishi bo‘yicha snaryadning (bizning holda ^{16}O) tezligiga teng bo‘lsa, sodir bo‘ladi. Nishon ^{54}Fe dastlab tinch holatda. Reaksiya natijasida ^{58}Ni yuqori qo‘zg‘algan holatlaridan biriga o‘tadi. Agar ^{16}O snaryadining kinetik energiyasi 50 MeV bo‘lsa, qo‘zg‘alish energiyasini toping (natijani MeV da ifodalang). Yorug‘lik tezligi $c = 3 \cdot 10^8$ m/s. *(2.2 ball)*

1.	$M(^{16}\text{O}) = 15,99491$ a.m.u.
2.	$M(^{54}\text{Fe}) = 53,93962$ a.m.u.
3.	$M(^{12}\text{C}) = 12,00000$ a.m.u.
4.	$M(^{58}\text{Ni}) = 57,93535$ a.m.u.

Table 1: 1-jadval. Reaksiyaga kirishuvchi yadrolarning asosiy holatdagi tinch massalari. 1 a.m.u. = 1,6605$\cdot 10^{-27}$ kg.

2. a) qismida muhokama qilingan qo‘zg‘algan holatda hosil bo‘lgan ^{58}Ni yadrosi o‘z harakati yo‘nalishida gamma-foton chiqarib, asosiy holatga o‘tadi. Ushbu yemirilishni ^{58}Ni tinch turgan sanoq sistemasida ko‘rib, foton chiqqandan keyin ^{58}Ni ning orqaga tepish energiyasini (ya’ni yadroning olgan kinetik energiyasini) toping. Shu sistemadagi foton energiyasi qancha? Laboratoriya sanoq sistemasida (ya’ni ^{58}Ni harakatlanayotgan yo‘nalishda joylashtirilgan detektor tomonidan o‘lchanadigan) foton energiyasi qancha bo‘ladi? *(1.6 ball)*

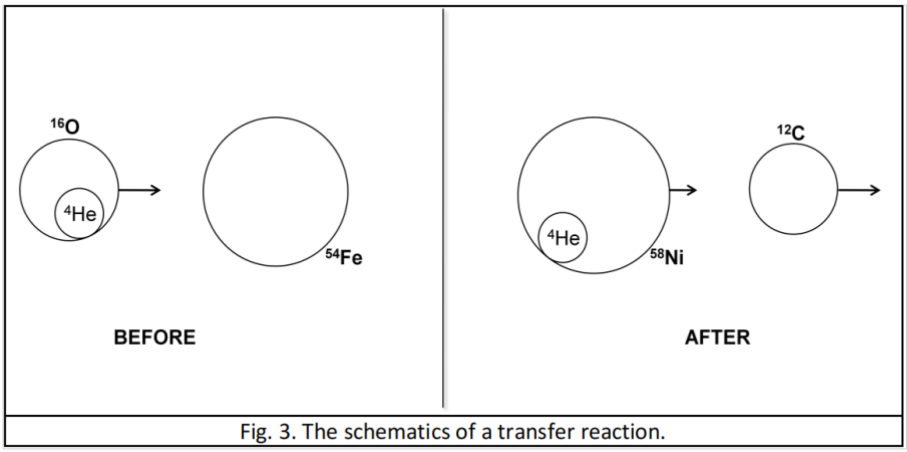

Fig. 3. The schematics of a transfer reaction.

Figure 8: 3-rasm. Transfer reaksiyasining sxemasi.

EKSPERIMENTAL MASALA 1: QATLAMLARNING ELASTIKLIGI

Kirish

Prujinalar elastik materiallardan tayyorlangan bo‘lib, ular mexanik energiyani saqlash uchun ishlatilishi mumkin. Eng mashhur spiral prujinalar Guk qonuni bilan yaxshi tavsiflanadi: prujinaning itarish kuchi muvozanat uzunligidan siljishga chiziqli proporsional: $F = -k\Delta x$, bunda k – prujina qattiqligi, Δx – muvozanatdan siljish, F – kuch [1(a)-rasm]. Biroq, elastik prujinalar odatdagi spiral prujinalardan butunlay boshqa shaklga ega bo‘lishi mumkin va katta deformatsiyalar uchun Guk qonuni umuman qo‘llanilmaydi. Ushbu masalada biz elastik material qatlamidan yasalgan prujinaning xossalarini o‘lchaymiz, u 1(b)-rasmda sxematik tasvirlangan.

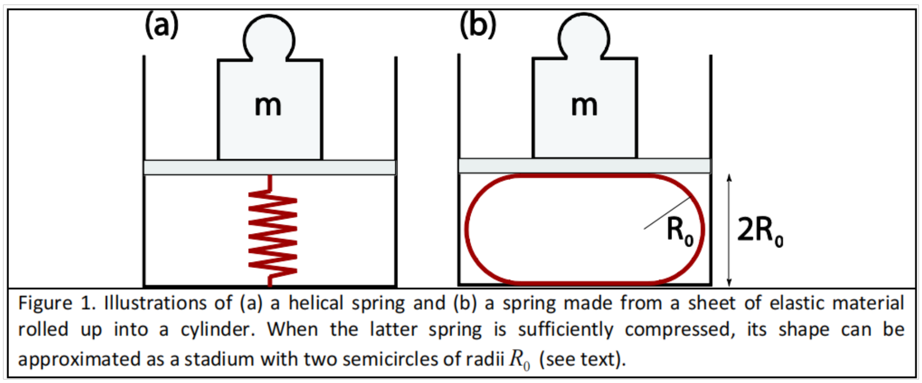

Figure 1. Illustrations of (a) a helical spring and (b) a spring made from a sheet of elastic material rolled up into a cylinder. When the latter spring is sufficiently compressed, its shape can be approximated as a stadium with two semicircles of radii R_0 (see text).

Figure 9: 1-rasm. (a) Spiral prujina va (b) elastik qatlamdan silindrga o‘ralgan prujina. Keyingi prujina yetarlicha siqilganda, uning shaklini radiusi R_0 bo‘lgan ikkita yarim aylana bilan stadion shaklida yaqinlashtirish mumkin.

Silindrik prujinaga o‘ralgan shaffof qatlam

Elastik material qatlamini (masalan, shaffof qatlam) olib, uni bukaylik. Qancha ko‘p buksak, qatlamda shuncha ko‘p elastik energiya saqlanadi. Elastik energiya qatlamning egriligiga bog‘liq. Egriligi katta bo‘lgan qatlam qismlari ko‘proq energiya saqlaydi (qatlamning tekis qismlari energiya saqlamaydi, chunki ularning egriligi nolga teng). Ushbu eksperimentda ishlatiladigan prujinalar to‘g‘ri burchakli shaffof qatlamlardan silindrlarga o‘ralgan holda tayyorlanadi (2-rasm). Silindrda saqlanadigan elastik energiya

$$E_{el} = \frac{\kappa}{2} \frac{1}{R_c^2} A,$$

bunda A – silindr yon sirtining maydoni (asoslaridan tashqari), R_c – uning radiusi, va κ parametri (egilish qattiqligi deb ataladi) materialning elastik xossalari va qatlam qalinligi bilan aniqlanadi. Bu yerda qatlamning cho‘zilishini e’tiborsiz qoldiramiz.

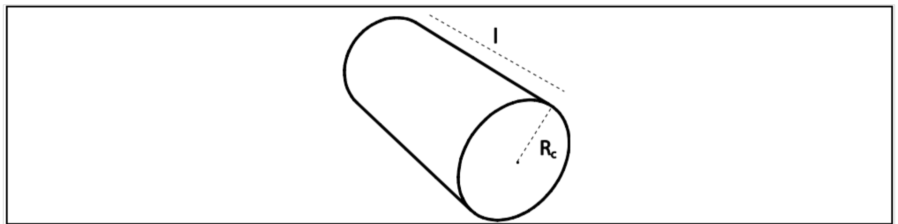

Figure 2. A schematic picture of an elastic sheet rolled into a cylinder of radius R_c and length l.

Figure 10: 2-rasm. Elastik qatlamning R_c radius va l uzunlikdagi silindrga o‘ralgan sxematik tasviri.

Faraz qilaylik, bunday silindr 1(b)-rasmdagidek siqilgan. Berilgan bosish kuchi F uchun muvozanatdan siljish shaffof qatlamning elastikligiga bog‘liq. Siqish kuchlarining ma’lum oralig‘ida siqilgan shaffof qatlamning shaklini stadion shakli bilan yaqinlashtirish mumkin. Stadion ko‘ndalang kesimi ikkita to‘g‘ri chiziq va ikkita yarim aylanadan (ikkalasi ham radiusi R_0) iborat. Siqilgan tizimning energiyasi minimal bo‘lishi shartidan

$$R_0^2 = \frac{l\kappa\pi}{2F}.$$

Kuch massani o‘lchaydigan masshtab (tarozi) bilan o‘lchanadi: $F = mg$, $g = 9,81\,\mathrm{m/s^2}$.

Eksperimental qurilma (1-masala)

Stolingizda (1-masala uchun) quyidagi buyumlar turibdi:

1. Press (tosh blok bilan birga);
2. Tarozi (5000 g gacha o‘lchaydi, TARA funksiyasi mavjud);
3. Shaffof qatlamlar (barcha qatlamlar 21 sm x 29,7 sm; ko‘k qatlam qalinligi 200 µm, rangsiz qatlam qalinligi 150 µm);
4. Yelimlovchi lenta (skotch);
5. Qaychi;
6. O‘lchagich;
7. To‘g‘ri burchakli yog‘och plastinka (plastinka tarozi ustiga qo‘yiladi, qatlam esa plastinka ustiga joylashtiriladi).

Qurilma 3-rasmdagidek ishlatiladi. Pressning yuqori plastinkasini gayka yordamida pastga va yuqoriga harakatlantirish mumkin, press tomonidan qo'llaniladigan kuch (massa) tarozi bilan o'lchanadi. **Muhim:** Gayka 360° aylantirilganda 2 mm ga harakatlanadi. (Kichik alyuminiy sterjen 1-eksperimentda ishlatilmaydi.)

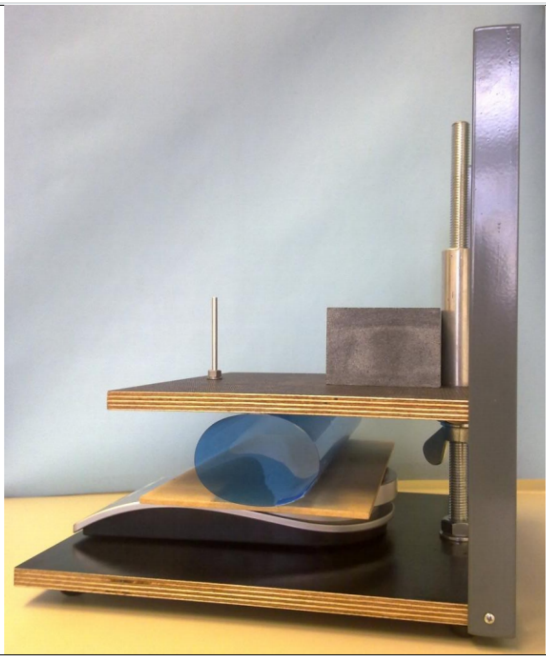

Figure 3. The photo of a setup for measuring the bending rigidity.

Figure 11: 3-rasm. Eksperimental qurilma.

Topshiriqlar

1. Ko'k qatlamlarni silindrlarga aylantiring: birini uzun tomoni bo'ylab, ikkinchisini qisqa tomoni bo'ylab; ularni yopishqoq lenta bilan mahkamlang. Qatlamning bir-biriga yopishgan qismi taxminan 0,5 sm bo'lsin.

   (a) Ikki silindrning har biri uchun tarozi ko'rsatgan massaning press plastinkalari orasidagi masofaga bog'liqligini o'lchang. *(1.9 ball)*

   (b) O'lchovlaringizni mos grafiklarda chizing. O'lchagich va ko'z bilan chiziqlar o'tkazing va har bir silindr uchun egilish qattiqligi κ ni aniqlang. Stadium yaqinlashishi amal qiladigan sohani belgilang.

   Stadium yaqinlashishi amal qiladigan R_0/R_c nisbatining pastki chegarasini baholang; bu yerda R_c yuksiz silindr(lar)ning radiusi. *(4.3 ball)*

   Natijalarning xato tahlili talab qilinmaydi.

2. Bitta rangsiz shaffof qatlamning egilish qattiqligini o'lchang. *(2.8 ball)*

3. Egilish qattiqligi κ izotrop materialning Young moduli Y va qatlam qalinligi d ga quyidagicha bog'liq:

$$\kappa = \frac{Yd^3}{12(1-\nu^2)},$$

bunda ν – materialning Puasson koeffitsienti; koʻpchilik materiallar uchun $\nu \approx 1/3$. Oldingi oʻlchovlardan koʻk va rangsiz shaffof qatlamlarning Young modulini aniqlang. *(1.0 ball)*

EKSPERIMENTAL MASALA 2: MAGNITLAR ORASIDAGI KUCHLAR, BARQARORLIK VA SIMETRIYA TUSHUNCHALARI

Kirish

S maydonli konturda aylanayotgan I tok $m = IS$ kattalikdagi magnit moment hosil qiladi [1(a)-rasm]. Doimiy magnitni temirning (Fe) kichik magnit momentlari to'plami deb qarash mumkin, ularning har biri tokli konturning magnit momentiga o'xshash. Magnitning bunday (Amper) modeli 1(b)-rasmda tasvirlangan. Umumiy magnit moment barcha kichik magnit momentlarning yig'indisiga teng va u janubiy qutbdan shimoliy qutbga yo'nalgan.

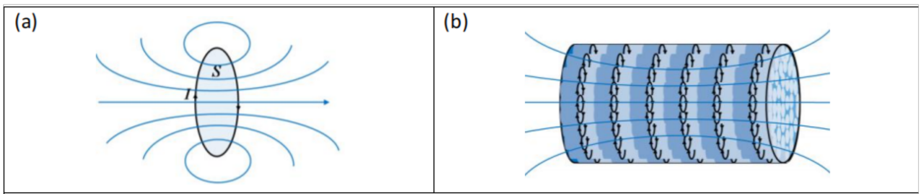

Figure 1. (a) Illustration of a current loop and the produced magnetic field. (b) Ampère's model of a magnet as a collection of small current loops.

Figure 12: 1-rasm. (a) Tokli kontur va uning hosil qilgan magnit maydoni. (b) Magnitning Amper modeli – kichik tokli konturlar to'plami.

Magnitlar orasidagi kuchlar

Ikki magnit orasidagi kuchni hisoblash nazariy jihatdan oddiy masala emas. Ma'lumki, ikki magnitning bir xil qutblari bir-birini itaradi, qarama-qarshi qutblari esa tortadi. Ikki tokli kontur orasidagi kuch ulardagi toklarning kuchiga, ularning shakliga va o'zaro masofaga bog'liq. Agar konturlardan biridagi tokni teskari tomonga o'zgartirsak, ular orasidagi kuch kattaligi jihatidan bir xil, ammo yo'nalishi qarama-qarshi bo'ladi.

Ushbu masalada siz ikki magnit – halqa-magnit va sterjen-magnit – orasidagi kuchlarni eksperimental ravishda tadqiq qilasiz. Bizni ikkala magnitning simmetriya o'qlari ustma-ust tushadigan geometriya qiziqtiradi (2-rasm). Sterjen-magnit z o'qi bo'ylab chapdan, halqa-magnit orqali va o'ngga qarab harakatlana oladi. Boshqa topshiriqlar qatori, magnitlar orasidagi kuchning z ga bog'liqligini o'lchashingiz so'raladi. $z = 0$ kelib chiqishi magnitlarning markazlari ustma-ust tushgan holga mos keladi.

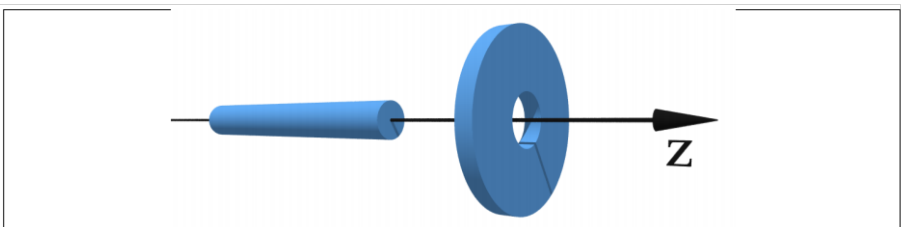

Figure 2. The rod- and the ring-magnet are aligned. The force between them changes as the rod-magnet moves along the z - axis.

Figure 13: 2-rasm. Halqa-magnit va sterjen-magnitning simmetriya o'qlari ustma-ust tushgan geometriya.

Sterjen-magnitning simmetriya o'qi (z-o'qi) bo'ylab harakatlanishini ta'minlash uchun halqa-magnit shaffof silindr ichiga mustahkam o'rnatilgan bo'lib, uning z-o'qi bo'ylab tor teshik ochilgan. Shunday qilib, sterjen-magnit teshik orqali z-o'qi bo'ylab harakatlanishga majburlangan (3-rasm). Magnitlarning magnitlanishi z-o'qi bo'ylab yo'nalgan. Teshik magnitlarning radial barqarorligini ta'minlaydi.

Figure 3. Photo of two magnets and a transparent hollow cylinder; the rod-magnet moves through the cylinder's hole.

Figure 14: 3-rasm. Ikki magnit va shaffof ichi bo'sh silindr; sterjen-magnit silindr teshigi orqali harakatlanadi.

Eksperimental qurilma (2-masala)

Stolingizda (2-masala uchun) quyidagi buyumlar turibdi:

1. Press (tosh blok bilan birga); agar kerak bo'lsa, alohida ko'rsatmalarga qarang.
2. Tarozi (5000 g gacha o'lchaydi, TARA funksiyasi mavjud); alohida ko'rsatmalarga qarang.
3. Yoniga halqa-magnit o'rnatilgan shaffof ichi bo'sh silindr.
4. Bitta sterjen-magnit.
5. Bitta tor yog'och tayoqcha (sterjen-magnitni silindrdan chiqarish uchun ishlatilishi mumkin).

Qurilma magnitlar orasidagi kuchlarni o'lchash uchun 4-rasmdagidek ishlatiladi. Pressning yuqori plastinkasini birinchi eksperimental masalaga nisbatan teskari tomonga burish kerak. Tor alyuminiy sterjen sterjen-magnitni ichi bo'sh silindr orqali bosish uchun ishlatiladi. Tarozi kuchni (massani) o'lchaydi. Pressning yuqori plastinkasini gayka yordamida pastga va yuqoriga harakatlantirish mumkin. **Muhim:** Gayka 360° aylantirilganda 2 mm ga harakatlanadi.

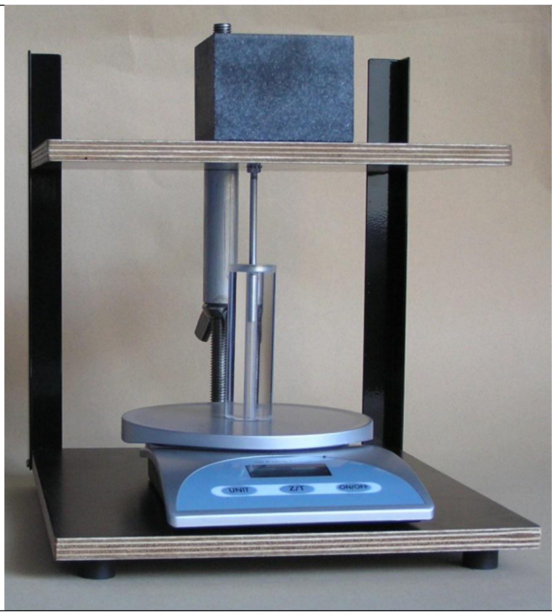

Figure 4. Photograph of the setup, and the way it should be used for measuring the force between the magnets.

Figure 15: 4-rasm. Qurilmaning fotosurati va magnitlar orasidagi kuchni oʻlchashda undan foydalanish usuli.

Topshiriqlar

1. z-oʻqi 2-rasmdagidek gorizontal joylashgan deb faraz qilib, ikki magnit orasidagi barcha muvozanat holatlarini sifat jihatidan aniqlang va ularni javoblar varaqasiga chizing. Muvozanat holatlarini barqaror (S) / beqaror (U) deb belgilang va bir xil qutblarni javoblar varaqasida koʻrsatilgan bir barqaror holat uchun misoldagidek soya bilan belgilang. Bu topshiriqni qoʻl va yogʻoch tayoqcha yordamida bajarishingiz mumkin. *(2.5 ball)*

2. Eksperimental qurilmadan foydalanib, ikki magnit orasidagi kuchni z-koordinatasi funktsiyasi sifatida oʻlchang. z-oʻqining musbat yoʻnalishi shaffof silindr ichiga qarab yoʻnalgan boʻlsin (kuch musbat yoʻnalishda boʻlsa, u musbat hisoblanadi). Magnit momentlari parallel boʻlgan konfiguratsiya uchun magnit kuchini $F_{\uparrow\uparrow}(z)$, antiparallel boʻlgan konfiguratsiya uchun esa $F_{\uparrow\downarrow}(z)$ bilan belgilang. **Muhim:** Sterjen-magnit massasini e'tiborsiz qoldiring (ya'ni gravitatsiyani hisobga olmang) va egri chiziqlarning turli qismlarini oʻlchash uchun magnitlar orasidagi kuchlarning simmetriyalaridan foydalaning. Agar kuchlarda biror simmetriyani topsangiz, ularni javoblar varaqasiga yozing. Oʻlchovlaringizni javoblar varaqasiga yozing; har bir jadval yoniga jadvalga mos keladigan magnitlar konfiguratsiyasini sxematik ravishda chizing (misol keltirilgan). *(3.0 ball)*

3. 2-topshiriqdagi oʻlchovlardan foydalanib, millimetrli qogʻozda $F_{\uparrow\uparrow}(z)$ ning $z > 0$ uchun batafsil funksional bogʻliqligini chizing. $F_{\uparrow\uparrow}(z)$ va $F_{\uparrow\downarrow}(z)$ egri chiziqlarining shakllarini (musbat va manfiy z-oʻqlari boʻylab) sxematik ravishda chizing. Har bir sxematik grafikda barqaror muvozanat nuqtalarining oʻrnini belgilang va magnitlarning mos konfiguratsiyasini chizing (1-topshiriqdagi kabi). *(4.0*

ball)

4. Agar sterjen-magnitning massasini e'tiborsiz qoldirmasak va z-o'qi vertikal joylashtirilsa, sifat jihatidan yangi barqaror muvozanat holatlari paydo bo'ladimi? Agar shunday bo'lsa, ularni 1-topshiriqdagi kabi javoblar varaqasiga chizing. *(0.5 ball)*

1-topshiriq yechimi

1a-topshiriq:

Metall shar asoslangan (grounded) bo'lgani uchun uning potensiali nolga teng: $V = 0$.

1b-topshiriq:

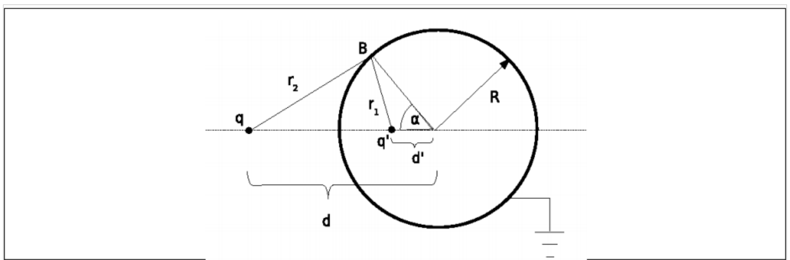

Fig 1. The potential at point *B* is zero.

.

Shar sirtidagi ixtiyoriy B nuqtani qaraylik (1-rasm). B nuqtaning q' zaryadidan masofasi

$$r_1 = \sqrt{R^2 + d'^2 - 2Rd' \cos\alpha},$$

B nuqtaning q zaryadidan masofasi esa

$$r_2 = \sqrt{R^2 + d^2 - 2Rd \cos\alpha}.$$

B nuqtadagi elektr potensiali

$$V = \frac{1}{4\pi\varepsilon_0} \left(\frac{q}{r_2} + \frac{q'}{r_1} \right).$$

Bu potensial nolga teng bo'lishi kerak:

$$\frac{q}{r_2} + \frac{q'}{r_1} = 0.$$

(1), (2) va (3) ni birlashtirib,

$$R^2 + d^2 - 2Rd\cos\alpha = \left(\frac{q}{q'} \right)^2 (R^2 + d'^2 - 2Rd' \cos\alpha).$$

Shar sirti ekvipotensial bo'lishi kerak, shuning uchun bu shart har bir α burchak uchun bajarilishi kerak. Bundan

$$d^2 + R^2 = \left(\frac{q}{q'} \right)^2 (R^2 + d'^2), \qquad dR = \left(\frac{q}{q'} \right)^2 (d'R).$$

Ushbu tenglamalarni yechib, q' zaryadining shar markazidan masofasi

$$d' = \frac{R^2}{d}$$

va zaryadning kattaligi

$$q' = -q\frac{R}{d}$$

kelib chiqadi.

1c-topshiriq:

q zaryadiga ta'sir qiluvchi kuchning kattaligi

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q^2 Rd}{(d^2 - R^2)^2}.$$

Bu kuch **tortishuvchi** (attractive) hisoblanadi.

2-topshiriq yechimi

2a-topshiriq:

A nuqtadagi elektr maydoni quyidagicha:

2a-topshiriq: A nuqtadagi elektr maydoni

$$\vec{E}_A = \left(\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} - \frac{1}{4\pi\varepsilon_0}\frac{q\frac{R}{d}}{(r - d + \frac{R^2}{d})^2}\right)\hat{r}.$$

2b-topshiriq: Juda katta r masofalar uchun $(1+a)^{-2} \approx 1 - 2a$ yaqinlashishini qo'llab,

$$\vec{E}_A = \frac{1}{4\pi\varepsilon_0}\frac{\left(1 - \frac{R}{d}\right)q}{r^2}\hat{r} - \frac{1}{4\pi\varepsilon_0}\frac{2q\frac{R}{d}\left(d - \frac{R^2}{d}\right)}{r^3}\hat{r}.$$

Umuman olganda, asoslangan metall shar nuqtaviy q zaryadini to'liq ekranlay olmaydi (hatto elektr maydoni $1/r^2$ dan tezroq kamayadigan ma'noda ham) va elektr maydonining r masofaga bog'liqligida asosiy had oddiy Kulon qonunidagi kabi bo'ladi.

2c-topshiriq: $d \to R$ chegarasida A nuqtadagi elektr maydoni nolga intiladi va asoslangan metall shar nuqtaviy zaryadni to'liq ekranlaydi.

3-topshiriq yechimi

3a-topshiriq: 2-rasmdagi konfiguratsiyani qaraylik. q zaryadining shar markazidan masofasi

$$d = \sqrt{l^2 + L^2 - 2lL\cos\alpha}.$$

q zaryadiga ta'sir qiluvchi elektr kuchining kattaligi

$$F = \frac{1}{4\pi\varepsilon_0}\frac{qq'}{(d-d')^2} = \frac{1}{4\pi\varepsilon_0}\frac{q^2 Rd}{(d^2 - R^2)^2},$$

bundan

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q^2 R\sqrt{l^2 + L^2 - 2lL\cos\alpha}}{(l^2 + L^2 - 2lL\cos\alpha - R^2)^2}.$$

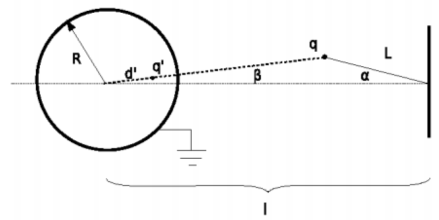

Fig 2. The pendulum formed by a charge near a grounded metallic sphere.

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3b-topshiriq: Elektr kuchi vektorining yo‘nalishi 3-rasmda tasvirlangan. α va β burchaklari

$$L\sin\alpha = d\sin\beta$$

bog‘lanishga ega, $\gamma = \alpha + \beta$. Ipga perpendikulyar kuch tashkil etuvchisi $F_\perp = F\sin\gamma$:

$$F_\perp = \frac{1}{4\pi\varepsilon_0}\frac{q^2R\sqrt{l^2+L^2-2lL\cos\alpha}}{(l^2+L^2-2lL\cos\alpha-R^2)^2}\sin(\alpha+\beta),$$

bunda

$$\beta = \arcsin\left(\frac{L}{\sqrt{L^2+l^2-2Ll\cos\alpha}}\sin\alpha\right).$$

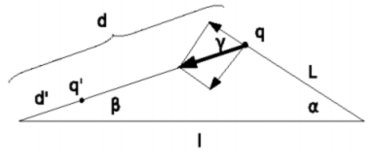


Fig 3. The direction of the force *F*.

.

3c-topshiriq: Matematik mayatnikning harakat tenglamasi:

$$mL\ddot{\alpha} = -F_\perp.$$

Kichik tebranishlar uchun α kichik. U holda $\beta \approx \alpha L/(l-L)$, $\gamma \approx l\alpha/(l-L)$ va $d \approx l - L$. Tenglamaga qo‘yib,

$$mL\frac{d^2\alpha}{dt^2} + \frac{1}{4\pi\varepsilon_0}\frac{q^2Rd}{(d^2-R^2)^2}\left(1+\frac{L}{d}\right)\alpha = 0.$$

Bu yerdan tebranish chastotasi

$$\omega = \frac{q}{d^2-R^2}\sqrt{\frac{Rd}{4\pi\varepsilon_0}\frac{1}{mL}\left(1+\frac{L}{d}\right)} = \frac{q}{(l-L)^2-R^2}\sqrt{\frac{Rl}{4\pi\varepsilon_0}\frac{1}{mL}}.$$

4-topshiriq yechimi

Avval zaryadlar to‘plamining elektrostatik energiyasi ta’rifiga asoslangan yechimni keltiramiz.

4a-topshiriq: Tizimning umumiy energiyasini tashqi zaryad va sferadagi induksiyalangan zaryadlar orasidagi elektrostatik o‘zaro ta’sir energiyasi $E_{el,1}$ va sferadagi zaryadlarning o‘zaro ta’sir energiyasi $E_{el,2}$ ga ajratish mumkin:

$$E_{el} = E_{el,1} + E_{el,2}.$$

Sferada induksiyalangan N ta q_j zaryadlari $\vec{r}_j$ nuqtalarda joylashgan. Tasvir zaryadining ta'rifiga ko'ra, sfera sirtidagi potensialni tasvir zaryadi hosil qiladi:

$$\frac{q'}{|\vec{r}-\vec{d}'|} = \sum_{j=1}^{N} \frac{q_j}{|\vec{r}_j - \vec{r}|},$$

bunda $\vec{r}$ – sfera sirtidagi nuqta, $\vec{d}'$ – tasvir zaryadining vektori. Sfera sirtida potensial nolga tengligidan

$$\frac{q'}{|\vec{r}-\vec{d}'|} + \frac{q}{|\vec{r}-\vec{d}|} = 0.$$

Tashqi zaryad va induksiyalangan zaryadlar orasidagi o'zaro ta'sir energiyasi:

$$\begin{aligned}
E_{el,1} &= \frac{q}{4\pi\varepsilon_0} \sum_{i=1}^{N} \frac{q_i}{|\vec{r}_i - \vec{d}|} \\
&= \frac{1}{4\pi\varepsilon_0} \frac{qq'}{|\vec{d}-\vec{d}'|} \\
&= \frac{1}{4\pi\varepsilon_0} \frac{q^2 R}{d^2 - R^2}.
\end{aligned}$$

4b-topshiriq: Sfera sirtidagi induksiyalangan zaryadlarning o'zaro ta'sir energiyasi:

$$\begin{aligned}
E_{el,2} &= \frac{1}{2} \cdot \frac{1}{4\pi\varepsilon_0} \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j\neq i}}^{N} \frac{q_i q_j}{|\vec{r}_i - \vec{r}_j|} \\
&= \frac{1}{2} \cdot \frac{1}{4\pi\varepsilon_0} \sum_{i=1}^{N} q_i \frac{q'}{|\vec{r}_i - \vec{d}'|} \\
&= \frac{1}{2} \cdot \frac{1}{4\pi\varepsilon_0} \sum_{i=1}^{N} q_i \frac{q}{|\vec{r}_i - \vec{d}|} \\
&= \frac{1}{2} \cdot \frac{1}{4\pi\varepsilon_0} \frac{qq'}{|\vec{d}-\vec{d}'|} \\
&= \frac{1}{2} \cdot \frac{1}{4\pi\varepsilon_0} \frac{q^2 R}{d^2 - R^2}.
\end{aligned}$$

Bu yerda ikkinchi qatorda (22) dan, uchinchi qatorda (23) dan, to'rtinchi qatorda esa yana (22) dan foydalanildi.

4c-topshiriq: $E_{el,1}$ va $E_{el,2}$ ni birlashtirib, elektrostatik o'zaro ta'sirning umumiy energiyasi:

$$E_{el}(d) = -\frac{1}{2} \cdot \frac{1}{4\pi\varepsilon_0} \frac{q^2 R}{d^2 - R^2}.$$

Muqobil yechim – bajarilgan ishning ta'rifidan kelib chiqadi. Integral

$$\int_d^{\infty} \frac{x\,dx}{(x^2 - R^2)^2} = \frac{1}{2} \cdot \frac{1}{d^2 - R^2}$$

dan foydalanib, q zaryadini cheksizlikdan d masofaga keltirish uchun bajarilgan ishni hisoblaymiz:

$$\begin{aligned} E_{el}(d) &= -\int_d^\infty F(\vec{x})d\vec{x} = \int_d^\infty F(\vec{x})d\vec{x} \\ &= \int_d^\infty \left(-\frac{1}{4\pi\varepsilon_0}\frac{q^2Rx}{(x^2-R^2)^2}\right)dx \\ &= -\frac{1}{2}\cdot\frac{1}{4\pi\varepsilon_0}\frac{q^2R}{d^2-R^2}. \end{aligned}$$

Bu 4c-topshiriqning yechimini beradi. q zaryadi va sfera orasidagi elektrostatik energiya, tasvir zaryadining ta'rifiga ko'ra, q va q' zaryadlari orasidagi energiyaga teng.

4-topshiriqning muqobil yechimidan:

$$E_{el,1} = \frac{1}{4\pi\varepsilon_0}\frac{qq'}{(d-d')} = -\frac{1}{4\pi\varepsilon_0}\frac{q^2R}{d^2-R^2}$$

Bu 4a-topshiriqni yechadi. Bundan darhol sferadagi zaryadlarning o'zaro elektrostatik energiyasi:

$$E_{el,2} = \frac{1}{2}\cdot\frac{1}{4\pi\varepsilon_0}\frac{q^2R}{d^2-R^2}$$

Bu 4b-topshiriqni yechadi.

YECHIM – MO‘RI FIZIKASI

1-topshiriq yechimi

a) Mo‘rining samarali ishlashi uchun zarur minimal balandlik.

$p(z)$ – z balandlikdagi havo bosimi bo‘lsin. Farazga ko‘ra $p(z) = p(0) - \rho_{\text{Havo}} gz$, bunda $p(0)$ – dengiz sathidagi atmosfera bosimi. Mo‘ri bo‘ylab Bernulli qonuni:

$$\frac{1}{2}\rho_{\text{Tutun}} v(z)^2 + \rho_{\text{Tutun}} gz + p_{\text{Tutun}}(z) = \text{const.}$$

Bu tenglamani ikki nuqtada qo‘llaymiz: (i) o‘choqda $z = -\varepsilon$ (ε juda kichik musbat), (ii) mo‘ri tepasida $z = h$:

$$\frac{1}{2}\rho_{\text{Tutun}} v(h)^2 + \rho_{\text{Tutun}} gh + p_{\text{Tutun}}(h) \approx p_{\text{Tutun}}(-\varepsilon)$$

O‘choqdagi gaz tezligi ahamiyatsiz va $-\rho_{\text{Tutun}} g\varepsilon \approx 0$. Minimal balandlikda mo‘ri tepasidagi tutun bosimi $p_{\text{Tutun}}(h) \approx p(h)$ ga teng. O‘choqda $p_{\text{Tutun}}(-\varepsilon) \approx p(0)$. Bernulli tenglamasi:

$$\frac{1}{2}\rho_{\text{Tutun}} v(h)^2 + \rho_{\text{Tutun}} gh + p(h) \approx p(0)$$

Bundan

$$v(h) = \sqrt{2gh\left(\frac{\rho_{\text{Havo}}}{\rho_{\text{Tutun}}} - 1\right)}.$$

Mo‘ri samarali bo‘lishi uchun barcha gaz mahsulotlari atmosferaga chiqarilishi kerak:

$$v(h) \geq \frac{B}{A}.$$

Shuning uchun

$$h \geq \frac{B^2}{A^2}\frac{1}{2g}\frac{1}{\frac{\rho_{\text{Havo}}}{\rho_{\text{Tutun}}} - 1}.$$

O‘choqdagi tutunni ideal gaz deb hisoblaymiz (bosim $p(0)$, harorat T_{Tutun}). Havo bir xil harorat va bosimda bir xil zichlikka ega bo‘lar edi. Shuning uchun

$$\frac{\rho_{\text{Havo}}}{\rho_{\text{Tutun}}} = \frac{T_{\text{Tutun}}}{T_{\text{Havo}}}, \quad \text{va nihoyat}$$

$$h \geq \frac{B^2}{A^2}\frac{1}{2g}\frac{T_{\text{Havo}}}{T_{\text{Tutun}} - T_{\text{Havo}}} = \frac{B^2}{A^2}\frac{1}{2g}\frac{T_{\text{Havo}}}{\Delta T}.$$

Minimal balandlikda tenglik belgisi olinadi.

b) Issiq mintaqadagi mo‘rining balandligi.

$$\frac{h(30)}{h(-30)} = \frac{T(30)}{T_{\text{Tutun}} - T(30)} \cdot \frac{T_{\text{Tutun}} - T(-30)}{T(-30)}; \quad h(30) = 145\,\text{m}.$$

c) Gaz tezligining mo‘ri bo‘ylab o‘zgarishi. Tezlik o‘zgarmas:

$$v = \sqrt{2gh\left(\frac{T_{\text{Tutun}}}{T_{\text{Havo}}} - 1\right)} = \sqrt{2gh\frac{\Delta T}{T_{\text{Havo}}}}.$$

Bu uzluksizlik tenglamasidan kelib chiqadi (ρ_{Tutun} o'zgarmas). Gazlar o'choqdan mo'riga kirganda tezlik taxminan noldan shu o'zgarmas qiymatga sakraydi. Minimal balandlikda $v = B/A$.

d) Gaz bosimining mo'ri bo'ylab o'zgarishi. Berilgan z balandlikda Bernulli tenglamasidan:

$$p_{\text{Tutun}}(z) = p(0) - (\rho_{\text{Havo}} - \rho_{\text{Tutun}})gh - \rho_{\text{Tutun}}gz.$$

Bosim mo'riga kirishda keskin o'zgaradi.

2-topshiriq yechimi

a) Quyosh mo'ri elektr stansiyasining FIKi. Δt vaqt oralig'ida chiqarilgan issiq havoning kinetik energiyasi:

$$E_{\text{kin}} = \frac{1}{2}(Av\Delta t\rho_{\text{Issiq}})v^2 = Av\Delta t\rho_{\text{Issiq}}gh\frac{\Delta T}{T_{\text{Havo}}}.$$

Vaqt birligida mo'ridan chiqadigan havo massasi $w = Av\rho_{\text{Issiq}}$ bo'lsa, kinetik energiyaga mos keladigan quvvat:

$$P_{\text{kin}} = wgh\frac{\Delta T}{T_{\text{Havo}}}.$$

Havoni isitish uchun sarflangan quyosh quvvati:

$$P_{\text{Quyosh}} = GS = wc\Delta T.$$

Foydali ish koeffitsienti:

$$\eta = \frac{P_{\text{kin}}}{P_{\text{Quyosh}}} = \frac{gh}{cT_{\text{Havo}}}.$$

b) FIKning mo'ri balandligiga bog'liqligi chiziqli.

3-topshiriq yechimi (Manzanares prototipi)

a) FIK:

$$\eta = \frac{gh}{cT_{\text{Havo}}} = \frac{9,81 \cdot 195}{1012 \cdot 295} \approx 0,0064 = 0,64\%.$$

b) Quvvat:

$$P = GS\eta = G\left(\frac{D}{2}\right)^2\pi\eta = 150 \cdot (122)^2\pi \cdot 0,0064 \approx 45\,\text{kW}.$$

c) Bir quyoshli kunda (8 soat) ishlab chiqarilgan energiya:

$$E = P \cdot t = 45\,\text{kW} \cdot 8\,\text{soat} = 360\,\text{kWh}.$$

4-topshiriq yechimi

Havo massaviy oqim tezligi:

$$w = Av\rho_{\text{Issiq}} = A\sqrt{2gh\frac{\Delta T}{T_{\text{Havo}}}}\rho_{\text{Issiq}}.$$

Boshqa tomondan,

$$w = \frac{GS}{c\Delta T}.$$

Tenglashtirib, ΔT ni topamiz:

$$\Delta T = \left(\frac{G^2S^2T_{\text{Havo}}}{A^2c^2\rho_{\text{Issiq}}^2 \cdot 2gh}\right)^{1/3} \approx 9,1\,\text{K}.$$

Bundan massaviy oqim tezligi:

$$w = 760\,\text{kg/s}.$$

YECHIM – ATOM YADROSINING MODELI

1-topshiriq yechimi

a) SC tizimida berilgan kubning 8 burchagining har birida bitta birlik (atom, nuklon) mavjud, lekin u 8 ta qo‘shni kub tomonidan baham ko‘riladi – bu har bir kubga bitta nuklon to‘g‘ri keladi. Nuklonlar bir-biriga tegib turgan deb faraz qilsak, kub cheti uzunligi $a = 2r_N$. Bitta nuklonning hajmi:

$$V_N = \frac{4}{3}\pi r_N^3 = \frac{4}{3}\pi\left(\frac{a}{2}\right)^3 = \frac{\pi}{6}a^3,$$

bundan

$$f = \frac{V_N}{a^3} = \frac{\pi}{6} \approx 0,52.$$

b) Yadroning massa zichligi:

$$\rho_m = f\frac{m_N}{V_N} = 0,52 \cdot \frac{1,67 \cdot 10^{-27}}{\frac{4}{3}\pi(0,85 \cdot 10^{-15})^3} \approx 3,40 \cdot 10^{17}\,\frac{\text{kg}}{\text{m}^3}.$$

c) Proton va neytronlar soni taxminan teng deb olib, zaryad zichligi:

$$\rho_c = \frac{f}{2}\frac{e}{V_N} = \frac{0,52}{2} \cdot \frac{1,6 \cdot 10^{-19}}{\frac{4}{3}\pi(0,85 \cdot 10^{-15})^3} \approx 1,63 \cdot 10^{25}\,\frac{\text{C}}{\text{m}^3}.$$

Berilgan A nuklonli yadroning hajmi:

$$V = \frac{AV_N}{f}, \qquad R = r_N\left(\frac{A}{f}\right)^{1/3} = \frac{r_N}{f^{1/3}}A^{1/3} = \frac{0,85}{0,52^{1/3}}A^{1/3} = 1,06\,\text{fm} \cdot A^{1/3}.$$

Keyingi topshiriqlarda $r_0 = 1,06$ fm deb belgilanadi.

2-topshiriq yechimi

Sirt nuklonlarining sonini baholaymiz. Sirt nuklonlari yadro sirtida qalinligi $2r_N$ bo‘lgan sferik qobiqda joylashgan. Qobiq hajmi:

$$\begin{aligned} V_{\text{sirt}} &= \frac{4}{3}\pi R^3 - \frac{4}{3}\pi(R-2r_N)^3 \\ &= 8\pi R r_N(R-2r_N) + \frac{4}{3}\pi\cdot 8r_N^3 = 8\pi(R^2 r_N - 2Rr_N^2 + \frac{4}{3}r_N^3). \end{aligned}$$

Sirt nuklonlari soni:

$$\begin{aligned} A_{\text{sirt}} &= f\frac{V_{\text{sirt}}}{V_N} = f\frac{8\pi(R^2 r_N - 2Rr_N^2 + \frac{4}{3}r_N^3)}{\frac{4}{3}\pi r_N^3} \\ &= 6f\left(\left(\frac{R}{r_N}\right)^2 - 2\frac{R}{r_N} + \frac{4}{3}\right) \\ &= 6f\left(\left(\frac{A}{f}\right)^{2/3} - 2\left(\frac{A}{f}\right)^{1/3} + \frac{4}{3}\right) \\ &= 6f^{1/3}A^{2/3} - 12f^{2/3}A^{1/3} + 8f \\ &\approx 4,84A^{2/3} - 7,80A^{1/3} + 4,19. \end{aligned}$$

Bog‘lanish energiyasi:

$$\begin{aligned} E_b &= (A - A_{\text{sirt}})a_V + A_{\text{sirt}}\frac{a_V}{2} = Aa_V - A_{\text{sirt}}\frac{a_V}{2} \\ &= Aa_V - (3f^{1/3}A^{2/3} - 6f^{2/3}A^{1/3} + 4f)a_V \\ &= 15,8A - 38,20A^{2/3} + 61,58A^{1/3} - 33,09\ \text{MeV}. \end{aligned}$$

3-topshiriq yechimi – Kulon effektlari

a) $Q_0 = Ze$ ni qo‘llasak, yadroning elektrostatik energiyasi:

$$U_c = \frac{3(Ze)^2}{20\pi\varepsilon_0 R}.$$

Har bir proton o‘ziga emas, balki qolganlarga ta’sir qilishini hisobga olish uchun Z^2 ni $Z(Z-1)$ ga almashtiramiz:

$$U_c = \frac{3Z(Z-1)e^2}{20\pi\varepsilon_0 R}.$$

b) $R = r_N f^{-1/3}A^{1/3}$ ni qo‘yib: Elektrostatik energiya formulasida $R = r_N f^{-1/3}A^{1/3}$ ni qo‘yib:

$$\Delta E_b = -\frac{3e^2 f^{1/3}}{20\pi\varepsilon_0 r_N}\frac{Z(Z-1)}{A^{1/3}} = -\frac{Z(Z-1)}{A^{1/3}}\cdot 1,31\times 10^{-13}\ \text{J} = -\frac{Z(Z-1)}{A^{1/3}}\cdot 0,815\ \text{MeV} \approx -0,204A^{5/3}\ \text{MeV-}$$

bunda $Z \approx A/2$ ishlatilgan. Kulon itarishi bog‘lanish energiyasini kamaytiradi, shuning uchun asosiy had oldida manfiy ishora. Bog‘lanish energiyasining to‘liq formulasi:

$$E_b = Aa_V - 3f^{1/3}A^{2/3}a_V + 6f^{2/3}A^{1/3}a_V - 4fa_V - \frac{3e^2 f^{1/3}}{20\pi\varepsilon_0 r_N}\left(\frac{A^{5/3}}{4} - \frac{A^{2/3}}{2}\right).$$

4-topshiriq yechimi – Og‘ir yadrolarning bo‘linishi

a) Kinetik energiya ikkita kichik yadro va bitta katta yadroning bog‘lanish energiyalari farqi hamda ikkita kichik yadro (har biri $Z/2 = A/4$ nuklonli) orasidagi Kulon energiyasidan keladi:

$$\begin{aligned}
E_{\text{kin}}(d) &= 2E_b\left(\frac{A}{2}\right) - E_b(A) - \frac{1}{4\pi\varepsilon_0}\frac{A^2e^2}{4\cdot 4\cdot d} \\
&= -3f^{1/3}A^{2/3}a_V(2^{1/3}-1) + 6f^{2/3}A^{1/3}a_V(2^{2/3}-1) \\
&\quad - 4fa_V - \frac{3e^2f^{1/3}}{20\pi\varepsilon_0 r_N}\left(\frac{A^{5/3}}{4} - \frac{A^{2/3}}{2}\right)\left(\frac{2^{1/3}-1}{2}\right) \\
&\quad - \frac{1}{4\pi\varepsilon_0}\frac{A^2e^2}{16d},
\end{aligned}$$

(bunda birinchi had Aa_V qisqarib ketadi).

b) $d = 2R(A/2)$ bo‘lganda kinetik energiya:

$$\begin{aligned}
E_{\text{kin}} &= 2E_b\left(\frac{A}{2}\right) - E_b(A) - \frac{1}{4\pi\varepsilon_0}\frac{A^2e^2}{16\cdot 2r_NA^{1/3}f^{-1/3}} \\
&= -3f^{1/3}A^{2/3}a_V(2^{1/3}-1) + 6f^{2/3}A^{1/3}a_V(2^{2/3}-1) \\
&\quad - 4fa_V - \frac{e^2f^{1/3}}{4\pi\varepsilon_0 r_N}\left(\frac{3}{80}(2^{2/3}-1) + \frac{2^{1/3}}{128}\right)A^{5/3} \\
&\quad - \frac{e^2f^{1/3}}{4\pi\varepsilon_0 r_N}\left(\frac{3}{40}(2^{1/3}-1)\right)A^{2/3} \\
&= (0,02203A^{5/3} - 10,0365A^{2/3} + 36,175A^{1/3} - 33,091)\,\text{MeV}.
\end{aligned}$$

Raqamli hisoblar:

$$\begin{aligned}
A = 100: &\quad E_{\text{kin}} = -33,95\,\text{MeV}, \\
A = 150: &\quad E_{\text{kin}} = -30,93\,\text{MeV}.
\end{aligned}$$

$$\begin{aligned}
A = 200: &\quad E_{\text{kin}} = -14,10\,\text{MeV}, \\
A = 250: &\quad E_{\text{kin}} = +15,06\,\text{MeV}.
\end{aligned}$$

Bizning modelda bo‘linish $E_{\text{kin}}(d = 2R(A/2)) \geq 0$ bo‘lganda mumkin. Yuqoridagi raqamli baholardan ko‘rinadiki, bu taxminan $A = 200$ va $A = 250$ oralig‘ida sodir bo‘ladi – taxminiy baho $A \approx 225$. Quyidagi tenglamaning aniq raqamli hisobi:

$$E_{\text{kin}} = (0,02203A^{5/3} - 10,0365A^{2/3} + 36,175A^{1/3} - 33,091)\,\text{MeV} \geq 0$$

$A \geq 227$ uchun bo‘linish mumkin ekanligini ko‘rsatadi.

5-topshiriq yechimi – Transfer reaksiyalari

5a-topshiriq Bu qismni norelativistik yoki relativistik kinematika yordamida yechish mumkin.

Norelativistik yechim

Avval reaksiyada massaning energiyaga aylangan miqdorini (energiya ekvivalenti, Q-qiymat) topish kerak:

$$\begin{aligned}\Delta m &= (\text{reaksiyadan keyingi umumiy massa}) - (\text{reaksiyadan oldingi umumiy massa})\\ &= (57,93535 + 12,00000)\,\text{a.m.u.} - (53,93962 + 15,99491)\,\text{a.m.u.}\\ &= 0,00082\,\text{a.m.u.} = 1,3616 \cdot 10^{-30}\,\text{kg}.\end{aligned}$$

Eynshteynning massa-energiya ekvivalentligi formulasidan foydalanib:

$$\begin{aligned}Q &= (\text{reaksiyadan keyingi umumiy kinetik energiya}) - (\text{reaksiyadan oldingi umumiy kinetik energiya})\\ &= -\Delta m \cdot c^2 = -1,3616 \cdot 10^{-30} \cdot (2,99792458 \cdot 10^8)^2 = -1,2237 \cdot 10^{-13}\,\text{J}.\end{aligned}$$

1 MeV $= 1,602 \cdot 10^{-13}$ J ekanligini hisobga olib:

$$Q = -1,2237 \cdot 10^{-13}/1,602 \cdot 10^{-13} = -0,761\,\text{MeV}.$$

Endi bu masala energiya va impulsning saqlanish qonunlari yordamida yechiladi. Impulsning saqlanishi (bizni faqat ^{12}C va ^{16}O bir xil yo‘nalishga ega bo‘lgan hol qiziqtirgani uchun vektorlardan foydalanmasak ham bo‘ladi):

$$m(^{16}\text{O})\, v(^{16}\text{O}) = m(^{12}\text{C})\, v(^{12}\text{C}) + m(^{58}\text{Ni})\, v(^{58}\text{Ni}).$$

Energiyaning saqlanishi esa:

$$E_k(^{16}\text{O}) + Q = E_k(^{12}\text{C}) + E_k(^{58}\text{Ni}) + E_k(^{58}\text{Ni}).$$

bu yerda $E_x(^{58}\text{Ni})$ – ^{58}Ni ning uyg’onish energiyasi (yadroning barqaror holatdan yuqori energetik sathga o’tish energiyasi), va Q esa ushbu topshiriqning birinchi qismida hisoblangan. Ammo ^{12}C va ^{16}O bir xil tezlikka (harakat tezligiga) ega bo’lganligi sababli, impulsning saqlanish qonuni (tizimning to’qnashuvgacha va undan keyingi umumiy harakat miqdori tengligi) quyidagicha soddalashadi:

$$[m(^{16}\text{O}) - m(^{12}\text{C})]v(^{16}\text{O}) = m(^{58}\text{Ni})v(^{58}\text{Ni}) \tag{24}$$

Endi ^{58}Ni yadrosining kinetik energiyasini (harakat energiyasini) osongina topishimiz mumkin:

$$\begin{aligned}E_k(^{58}\text{Ni}) &= \frac{m(^{58}\text{Ni})v^2(^{58}\text{Ni})}{2} = \frac{[m(^{58}\text{Ni})v(^{58}\text{Ni})]^2}{2m(^{58}\text{Ni})} =\\ &= \frac{[[m(^{16}\text{O}) - m(^{12}\text{C})]v(^{16}\text{O})]^2}{2m(^{58}\text{Ni})} =\\ &= E_k(^{16}\text{O})\frac{[m(^{16}\text{O}) - m(^{12}\text{C})]^2}{m(^{58}\text{Ni})m(^{16}\text{O})}\end{aligned} \tag{25}$$

va nihoyat ^{58}Ni ning uyg’onish energiyasi (ichki energiya ortishi):

$$
\begin{aligned}
E_x(^{58}\text{Ni}) &= E_k(^{16}\text{O}) + Q - E_k(^{12}\text{C}) - E_k(^{58}\text{Ni}) = \\
&= E_k(^{16}\text{O}) + Q - \frac{m(^{12}\text{C})v^2(^{16}\text{O})}{2} - E_k(^{16}\text{O})\frac{[m(^{16}\text{O}) - m(^{12}\text{C})]^2}{m(^{58}\text{Ni})m(^{16}\text{O})} = \\
&= Q + E_k(^{16}\text{O}) - E_k(^{16}\text{O}) \cdot \frac{m(^{12}\text{C})}{m(^{16}\text{O})} - E_k(^{16}\text{O})\frac{[m(^{16}\text{O}) - m(^{12}\text{C})]^2}{m(^{58}\text{Ni})m(^{16}\text{O})} = \\
&= Q + E_k(^{16}\text{O})\left[1 - \frac{m(^{12}\text{C})}{m(^{16}\text{O})} - \frac{[m(^{16}\text{O}) - m(^{12}\text{C})]^2}{m(^{58}\text{Ni})m(^{16}\text{O})}\right] = \\
&= Q + E_k(^{16}\text{O})\frac{[m(^{16}\text{O}) - m(^{12}\text{C})] \cdot [m(^{58}\text{Ni}) - m(^{16}\text{O}) + m(^{12}\text{C})]}{m(^{58}\text{Ni})m(^{16}\text{O})}
\end{aligned}
\tag{26}
$$

E'tibor bering, suratdagi birinchi qavs ichidagi qiymat taxminan ko'chirilgan zarracha massasiga (^{4}He geliy yadrosi massasiga) teng, ikkinchi qavs esa taxminan nishon-yadro (to'qnashuv uchun mo'ljallangan yadro) ^{54}Fe massasiga teng. Sonli qiymatlarni o'rniga qo'yib, quyidagiga ega bo'lamiz:

$$
\begin{aligned}
E_x(^{58}\text{Ni}) &= -0.761 + 50 \cdot \frac{(15.99491 - 12.)(57.93535 - 15.99491 + 12.)}{57.93535 \cdot 15.99491} = \\
&= 10.866 \text{ MeV}
\end{aligned}
\tag{27}
$$

Relyativistik yechim (yorug'lik tezligiga yaqin yuqori tezliklar uchun hisoblash usuli)

Relyativistik talqinda yechim quyidagi tenglamalar juftligidan boshlab topiladi (birinchi tenglama – energiyaning saqlanish qonuni (to'liq relyativistik energiya saqlanishi), ikkinchisi esa impulsning saqlanish qonunidir (relyativistik harakat miqdori saqlanishi)):

$$
m(^{54}\text{Fe}) \cdot c^2 + \frac{m(^{16}\text{O}) \cdot c^2}{\sqrt{1 - v^2(^{16}\text{O})/c^2}} = \frac{m(^{12}\text{C}) \cdot c^2}{\sqrt{1 - v^2(^{12}\text{C})/c^2}} + \frac{m^*(^{58}\text{Ni}) \cdot c^2}{\sqrt{1 - v^2(^{58}\text{Ni})/c^2}}
\tag{28}
$$

$$
\frac{m(^{16}\text{O}) \cdot v(^{16}\text{O})}{\sqrt{1 - v^2(^{16}\text{O})/c^2}} = \frac{m(^{12}\text{C}) \cdot v(^{12}\text{C})}{\sqrt{1 - v^2(^{12}\text{C})/c^2}} + \frac{m^*(^{58}\text{Ni}) \cdot v(^{58}\text{Ni})}{\sqrt{1 - v^2(^{58}\text{Ni})/c^2}}
$$

Tenglamalardagi barcha massalar tinchlikdagi massalardir (zarrachaning harakatlanmayotgan holatdagi massasi); ^{58}Ni o'zining asosiy holatida (eng kichik energiyali barqaror holatida) EMAS, balki uyg'ongan holatlaridan (yuqori energiyali beqaror holatlaridan) biridadir (uning massasi m^* bilan belgilangan). ^{12}C va ^{16}O bir xil tezlikka (harakat tezligiga) ega bo'lganligi sababli, ushbu tenglamalar tizimi quyidagicha soddalashadi:

$$
\begin{aligned}
m(^{54}\text{Fe}) + \frac{m(^{16}\text{O}) - m(^{12}\text{C})}{\sqrt{1 - v^2(^{16}\text{O})/c^2}} &= \frac{m^*(^{58}\text{Ni})}{\sqrt{1 - v^2(^{58}\text{Ni})/c^2}} \\
\frac{(m(^{16}\text{O}) - m(^{12}\text{C})) \cdot v(^{16}\text{O})}{\sqrt{1 - v^2(^{16}\text{O})/c^2}} &= \frac{m^*(^{58}\text{Ni}) \cdot v(^{58}\text{Ni})}{\sqrt{1 - v^2(^{58}\text{Ni})/c^2}}
\end{aligned}
\tag{29}
$$

Ikkinchi tenglamani birinchisiga bo'lish quyidagicha natija beradi:

$$v(^{58}\mathrm{Ni}) = \frac{(m(^{16}\mathrm{O}) - m(^{12}\mathrm{C})) \cdot v(^{16}\mathrm{O})}{(m(^{16}\mathrm{O}) - m(^{12}\mathrm{C})) + m(^{54}\mathrm{Fe})\sqrt{1 - v^2(^{16}\mathrm{O})/c^2}} \tag{30}$$

Hujumkor zarrachaning (to'qnashuvni hosil qiluvchi uchuvchi zarrachaning) tezligini uning energiyasidan hisoblab topish mumkin:

$$\begin{aligned}
E_{kin}(^{16}\mathrm{O}) &= \frac{m(^{16}\mathrm{O}) \cdot c^2}{\sqrt{1 - v^2(^{16}\mathrm{O})/c^2}} - m(^{16}\mathrm{O}) \cdot c^2 \\
\sqrt{1 - v^2(^{16}\mathrm{O})/c^2} &= \frac{m(^{16}\mathrm{O}) \cdot c^2}{E_{kin}(^{16}\mathrm{O}) + m(^{16}\mathrm{O}) \cdot c^2} \\
v^2(^{16}\mathrm{O})/c^2 &= 1 - \left(\frac{m(^{16}\mathrm{O}) \cdot c^2}{E_{kin}(^{16}\mathrm{O}) + m(^{16}\mathrm{O}) \cdot c^2}\right)^2 \\
v(^{16}\mathrm{O}) &= \sqrt{1 - \left(\frac{m(^{16}\mathrm{O}) \cdot c^2}{E_{kin}(^{16}\mathrm{O}) + m(^{16}\mathrm{O}) \cdot c^2}\right)^2} \cdot c
\end{aligned} \tag{31}$$

Berilgan sonli qiymatlar uchun quyidagiga ega bo'lamiz:

$$\begin{aligned}
v(^{16}\mathrm{O}) &= \sqrt{1 - \left(\frac{15.99491 \cdot 1.6605 \cdot 10^{-27} \cdot (2.9979 \cdot 10^8)^2}{50 \cdot 1.602 \cdot 10^{-13} + 15.99491 \cdot (2.9979 \cdot 10^8)^2}\right)^2} \cdot c = \\
&= \sqrt{1 - 0.99666^2} \cdot c = 0.08172 \cdot c = 2.4498 \cdot 10^7 \text{ km/s}
\end{aligned} \tag{32}$$

Endi biz hisoblashimiz mumkin:

$$v(^{58}\mathrm{Ni}) = \frac{(15.99491 - 12.0) \cdot 2.4498 \cdot 10^7 \text{ km/s}}{(15.99491 - 12.0) + 53.93962\sqrt{1 - 0.08172^2}} = 1.6946 \cdot 10^6 \text{ km/s} \tag{33}$$

^{58}Ni ning uyg'ongan holatdagi massasi quyidagicha bo'ladi:

$$\begin{aligned}
m^*(^{58}\mathrm{Ni}) &= (m(^{16}\mathrm{O}) - m(^{12}\mathrm{C}))\frac{\sqrt{1 - v^2(^{58}\mathrm{Ni})/c^2}}{\sqrt{1 - v^2(^{16}\mathrm{O})/c^2}} \cdot \frac{v(^{16}\mathrm{O})}{v(^{58}\mathrm{Ni})} = \\
&= (15.99491 - 12.0)\frac{\sqrt{1 - (1.6945 \cdot 10^6/2.9979 \cdot 10^8)^2}}{\sqrt{1 - 0.08172^2}} \cdot \frac{2.4498 \cdot 10^7}{1.6945 \cdot 10^6} \text{ m.a.b.} = \\
&= 57.9470 \, \text{m.a.b.}
\end{aligned} \tag{34}$$

U holda ^{58}Ni ning uyg'onish energiyasi quyidagiga teng:

$$\begin{aligned}
E_x &= [m^*(^{58}\mathrm{Ni}) - m(^{58}\mathrm{Ni})] \cdot c^2 = (57.9470 - 57.93535) \cdot 1.6605 \cdot 10^{-27}(2.9979 \cdot 10^8)^2 = \\
&= 2.00722 \cdot 10^{-12}/1.602 \cdot 10^{-13} \text{ MeV/J} = 10.8636 \text{ MeV}
\end{aligned} \tag{35}$$

Relyativistik va norelyativistik natijalar 2 keV gacha aniqlikda o'zaro tengdir, shuning uchun ikkala usulni ham to'g'ri deb hisoblash mumkin – bundan kelib chiqadiki, berilgan dasta energiyasida relyativistik effektlar muhim ahamiyatga ega emas.

5b-topshiriq) Tinch turgan yadrodan gamma-kvant nurlanishi uchun energiya va impulsning saqlanish qonunlari quyidagini beradi:

$$E_x(^{58}\mathrm{Ni}) = E_\gamma + E_{\mathrm{recoil}} \qquad p_\gamma = p_{\mathrm{recoil}} \tag{36}$$

Gamma-nur va qaytgan (orqaga tepilgan) yadro, tabiiyki, qarama-qarshi yo'nalishlarga ega bo'ladi. Gamma-nur (foton) uchun energiya va impuls o'zaro quyidagicha bog'langan:

$$E_\gamma = p_\gamma \cdot c \tag{37}$$

a) qismda biz ushbu energiya oralig'ida yadroning harakati relyativistik emasligini ko'rib chiqdik, shuning uchun:

$$E_{\mathrm{recoil}} = \frac{p_{\mathrm{recoil}}^2}{2m(^{58}\mathrm{Ni})} = \frac{p_\gamma^2}{2m(^{58}\mathrm{Ni})} = \frac{E_\gamma^2}{2m(^{58}\mathrm{Ni})\cdot c^2} \tag{38}$$

Buni (36)-tenglamadagi energiyaning saqlanish qonuniga qo'yib, quyidagiga ega bo'lamiz:

$$E_x(^{58}\mathrm{Ni}) = E_\gamma + E_{\mathrm{recoil}} = E_\gamma + \frac{E_\gamma^2}{2m(^{58}\mathrm{Ni})\cdot c^2} \tag{39}$$

Bu esa quyidagi kvadrat tenglamaga keltiriladi:

$$E_\gamma^2 + 2m(^{58}\mathrm{Ni})c^2 \cdot E_\gamma + 2m(^{58}\mathrm{Ni})c^2 E_x(^{58}\mathrm{Ni}) = 0 \tag{40}$$

bu quyidagi yechimni beradi:

$$\begin{aligned} E_\gamma &= \frac{-2m(^{58}\mathrm{Ni})c^2 + \sqrt{4\left(m(^{58}\mathrm{Ni})c^2\right)^2 + 8m(^{58}\mathrm{Ni})c^2 E_x(^{58}\mathrm{Ni})}}{2} = \\ &= \sqrt{\left(m(^{58}\mathrm{Ni})c^2\right)^2 + 2m(^{58}\mathrm{Ni})c^2 E_x(^{58}\mathrm{Ni})} - m(^{58}\mathrm{Ni})c^2 \end{aligned} \tag{41}$$

Sonli qiymatlarni qo'yish quyidagini beradi:

$$E_\gamma = 10.8633\,\mathrm{MeV} \tag{42}$$

Sonli qiymatlarni qo'yishdan oldin (37)-tenglamani taqribiy tenglama ko'rinishiga keltirish ham mumkin:

$$E_\gamma = E_x\left(1 - \frac{E_x}{2m(^{58}\mathrm{Ni})c^2}\right) = 10.8633\,\mathrm{MeV} \tag{43}$$

Orqaga tepish energiyasi (recoil energy) endi osongina quyidagicha topiladi:

$$E_{\text{recoil}} = E_x(^{58}\text{Ni}) - E_\gamma = 1.1\,\text{keV} \tag{44}$$

Gamma-nurlanish tarqatayotgan yadro (^{58}Ni) katta tezlik bilan harakatlanayotganligi sababli, Doppler effekti tufayli gamma-nurining energiyasi o'zgaradi. Relativistik Doppler effekti (manba kuzatuvchi/detektor tomon harakatlanayotganda) ushbu formula bilan beriladi:

$$f_{\text{detector}} = f_{\gamma,\text{emitted}}\sqrt{\frac{1+\beta}{1-\beta}} \tag{45}$$

va foton energiyasi hamda chastotasi o'rtasida oddiy bog'lanish ($E = hf$) mavjud bo'lganligi sababli, energiya uchun ham shunga o'xshash ifodani olamiz:

$$E_{\text{detector}} = E_{\gamma,\text{emitted}}\sqrt{\frac{1+\beta}{1-\beta}} \tag{46}$$

bu yerda $\beta = v/c$ va v – nurlatgich (^{58}Ni yadrosi) tezligidir. ^{58}Ni tezligining hisoblangan qiymatini ((29)-tenglama) e'tiborga olsak, quyidagiga ega bo'lamiz:

$$E_{\text{detector}} = E_{\gamma,\text{emitted}}\sqrt{\frac{1+\beta}{1-\beta}} = 10.863\sqrt{\frac{1+0.00565}{1-0.00565}} = 10.925\,\text{MeV} \tag{47}$$

EKSPERIMENTAL MASALA 1 YECHIMI – ELASTIK QATLAMLAR

1-topshiriq (ko‘k qatlam, uzun tomon bo‘ylab o‘ralgan)

H

Solution: Exp. problem 1

Task 1		Points
(a)		0.95

m[g]	R_0[mm]
21	40.5
25	39.5
29	38.5
31	37.5
34	36.5
36	35.5
39	34.5
43	33.5
46	32.5
50	31.5
53	30.5
57	29.5
62	28.5
67	27.5
73	26.5
79	25.5
86	24.5
94	23.5
102	22.5
113	21.5
124	20.5
137	19.5
150	18.5
168	17.5
189	16.5
212	15.5
274	13.5
417	10.5

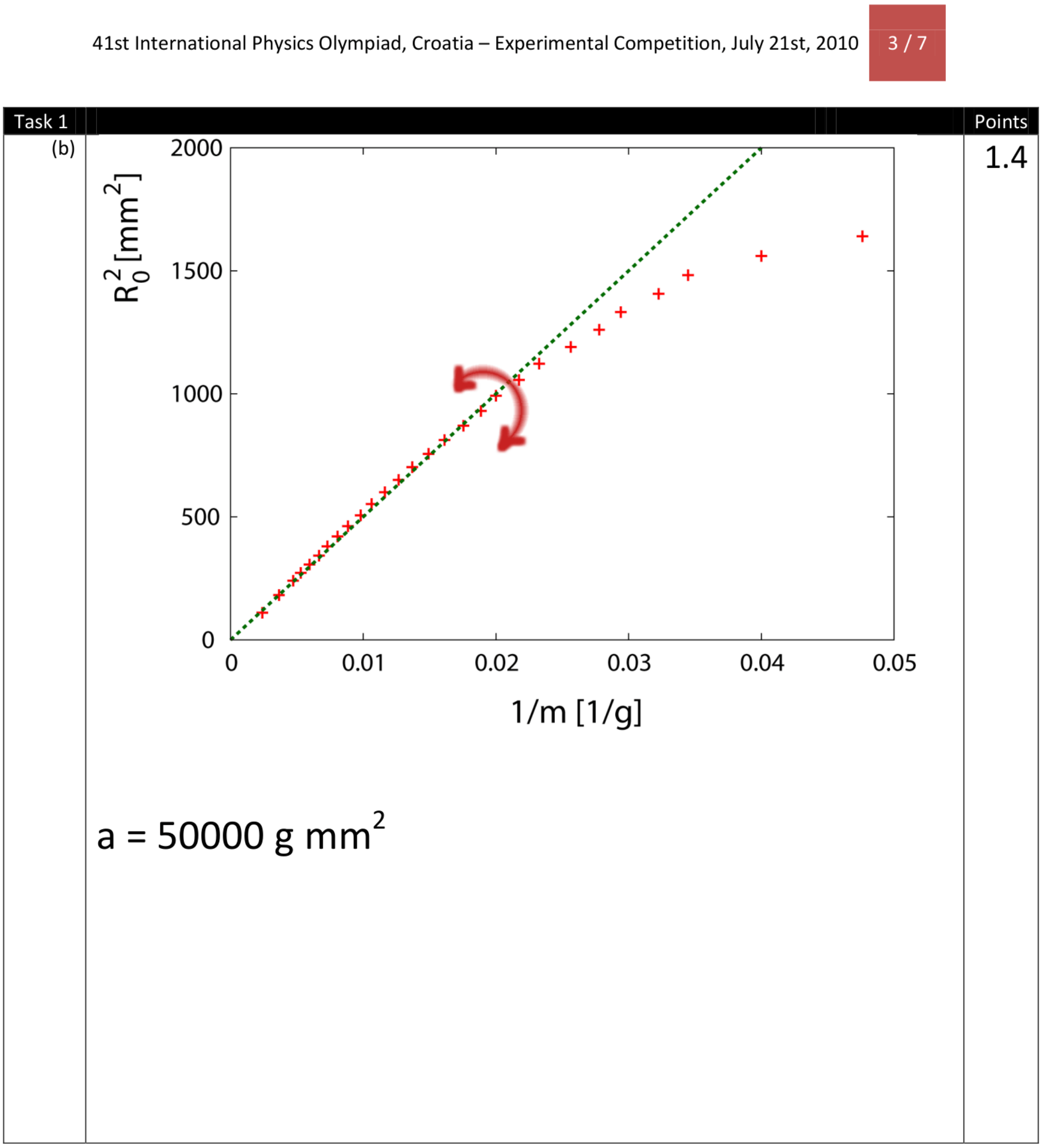
Figure 16: .

H

Task 1		Points
(a)		0.95

m[g]	R_0[mm]
40	29.9
42	29.8
45	29.6
47	29.4
50	29.3
52	29.1
54	28.9
57	28.8
59	28.6
61	28.4
64	28.3
71	27.8
78	27.3
92	26.3
105	25.3
118	24.3
129	23.3
143	22.3
157	21.3
171	20.3
189	19.3
211	18.3
235	17.3
259	16.3
293	15.3
336	14.3
386	13.3
449	12.3

Figure 17: .



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Task 1		Points
(b)	a = 50000 g mm^2	1.4

Figure 18: .

H

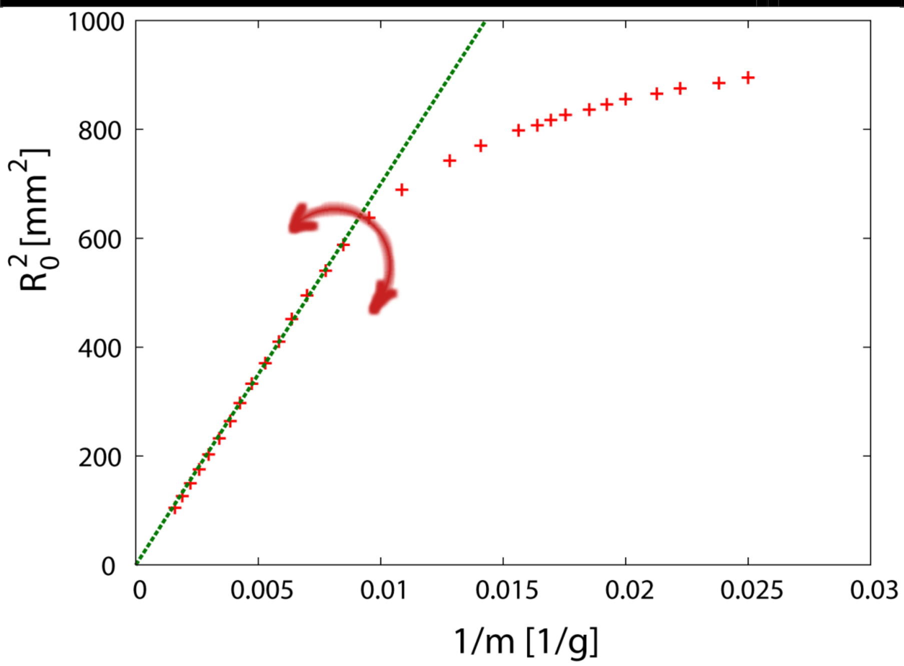
Task 1		Points
(b)	 $a = 70000\ g\ mm^2$	1.4

Figure 19: .

H

Task 1		Points
(b)	$\kappa = \frac{2ag}{\pi d} = 1.5\ \text{mJ}$	0.5
	$\kappa = \frac{2ag}{\pi d} = 1.5\ \text{mJ}$	0.5
	$\frac{R_0}{R_c} \leq 0.70$ $\frac{R_0}{R_c} \leq 0.77$	0.5

Task 2		Points
		0.9

m[g]	R_0[mm]
6	42.5
7	42.
9	41.
12	39.5
15	37.5
19	35.5
20	34.5
21	33.5
24	32.5
26	31.5
28	30.5
30	29.5
33	28.5
36	27.5
40	26.5
44	25.5
48	24.5
53	23.5
58	22.5
66	21.5
73	20.5
82	19.5
92	18.5
104	17.5
116	16.5
127	15.5
145	14.5
168	13.5

Figure 20: .

H

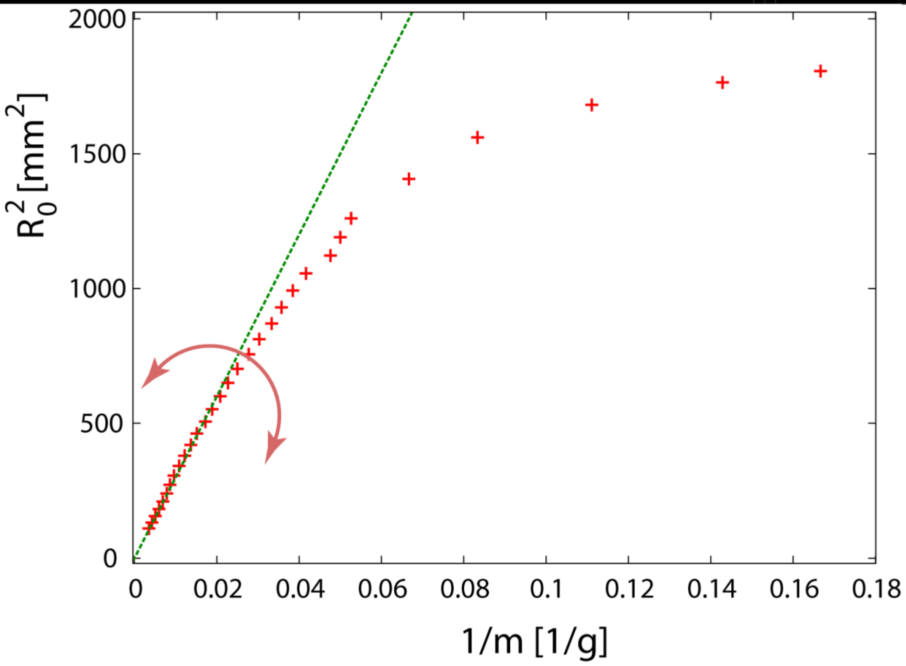
Task 2		Points
	 $a = 27000$ g mm^2	0.9

Figure 21: .

H

Task 2		Points
	$\kappa =$0.8 mJ	1.0
Task 3		**Points**
	Young modulus of the blue foil: Y=2.0 GPa Y=2.0 GPa	0.6
	Young modulus of the colorless foil: Y=2.5 GPa	0.4
Total:		10

Figure 22: .

H

Solution - Exp. Problem 2

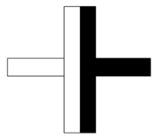
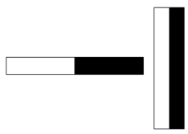
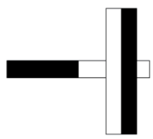
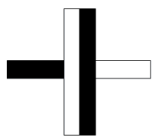
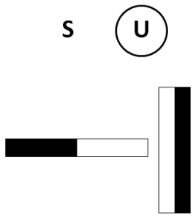
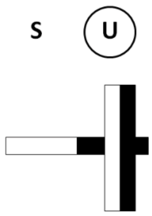
Task 1		Points
	S U 	0.25
	S U 	0.45
	S U 	0.45
	S U 	0.45

Figure 23: .



0.45



0.45

Task 2		Points

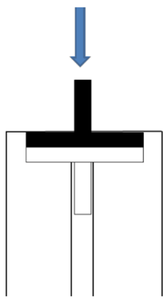
Symmetries that should be utilized in the measurements:

$F_{\uparrow\downarrow}(z) = -F_{\uparrow\downarrow}(-z)$
$F_{\uparrow\downarrow}(z) = -F_{\uparrow\uparrow}(z)$
From the two above one gets also $F_{\uparrow\uparrow}(z) = -F_{\uparrow\uparrow}(-z)$

0,6

By using the setup as it is, the whole curve can be measured by starting the measurements from three stable equilibrium points; the equilibrium point (z0) can be measured also by using the setup.

Configuration:



Measurements:

z0=0mm	m [g]	Δz [mm]
	0	0
	31	1
	55	2
	75	3
	97	4

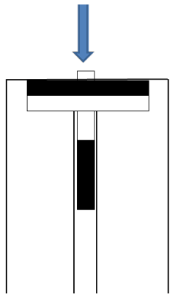
0,8

Figure 24: .

119	5
140	6
158	7
171	8
170	9
118	10
85	10,25
50	10,5
10	10,75

0,8

Configuration:



Measurements:

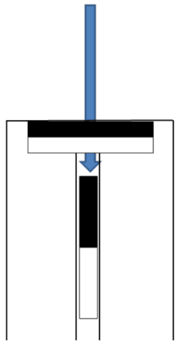
z0=10.8mm	m [g]	Δz [mm]
	0	0
	233	1
	538	2
	927	3
	996	3,5
	1124	4
	1154	4,5
	1213	5
	1212	5,5
	1120	6
	873	6,5
	284	7
	36	7,5

Figure 25: .

H

Configuration:

0,8



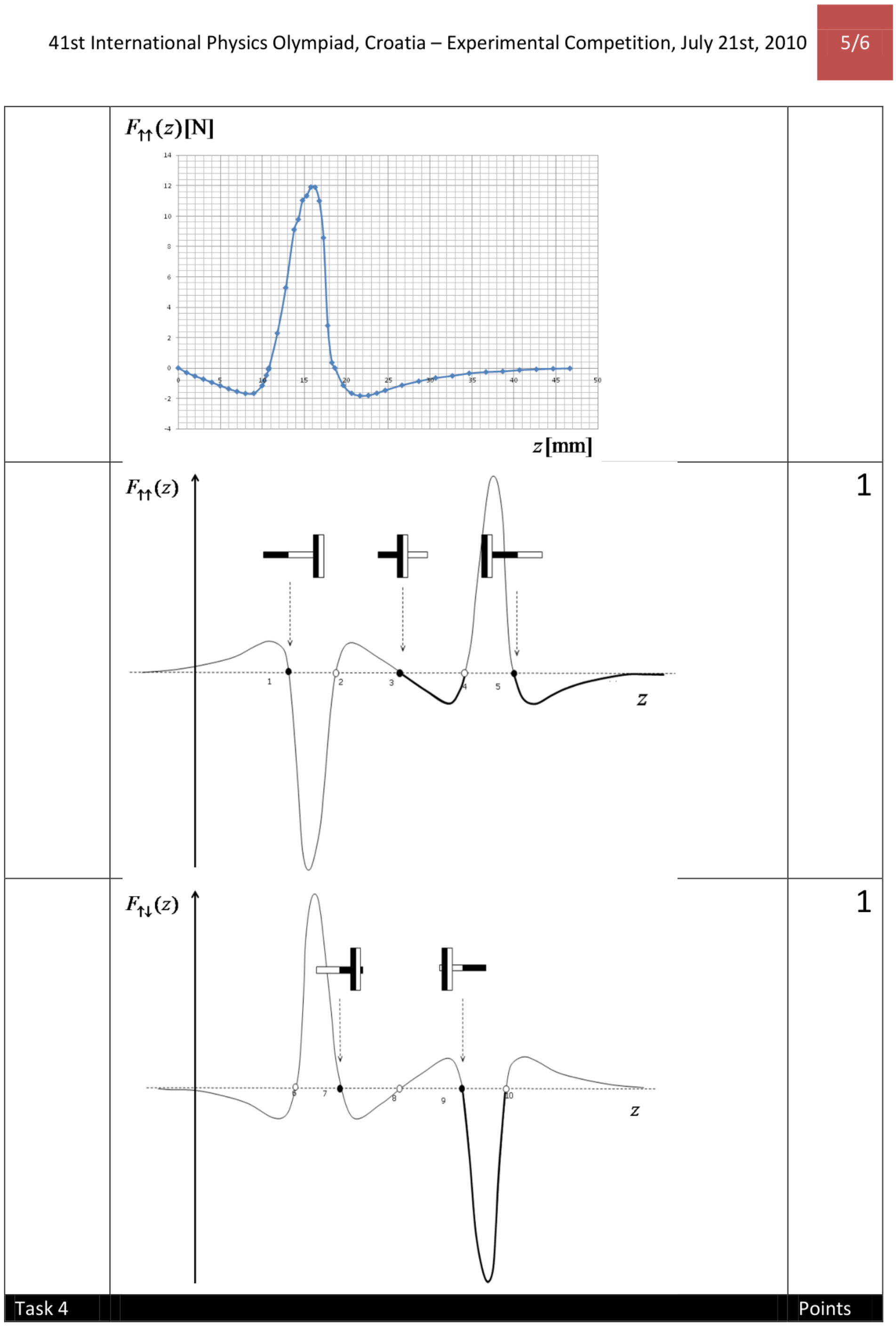
Measurements:

z0=18.6mm	m [g]	Δz [mm]
	0	0
	116	1
	170	2
	186	3
	184	4
	169	5
	150	6
	116	8
	89	10
	67	12
	53	14
	36	16
	27	18
	23	20
	14	22
	9	24
	5	26
	3	28

Task 3		Points
	Due to symmetry, it is sufficient to plot e.g., the following graph in detail:	2

Figure 26: .

H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5/6
$F_{\uparrow\uparrow}(z)$[N]
z[mm]
$F_{\uparrow\uparrow}(z)$
1
z
$F_{\uparrow\downarrow}(z)$
1
z
Task 4
Points

Figure 27: .

H

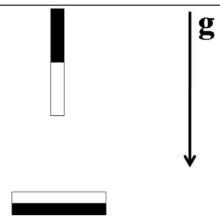
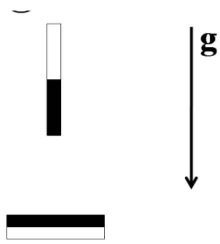
	 	0,5
	OR  	
Total:		10.0

Figure 28: .